Mortgage Interest Rate Risk and the Propagation of Shocks*

Michael Irwin† Matías Vieyra‡

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Abstract

Fixed-rate mortgages (FRMs) insure borrowers against changes in interest rates, but they do not eliminate interest-rate risk: they shift it to the financial intermediaries that hold the mortgages. We study how this shift affects the propagation of aggregate shocks in a heterogeneous-agent New Keynesian model with housing and long-term mortgages. The financial sector is described by a sufficient statistics approach that we discipline by targeting identified moments from the mortgage market. Without financial frictions, a monetary tightening reduces output 60% more over three years under variable-rate mortgages (VRMs) than under FRMs, because higher mortgage interest costs reduce borrowers' cash flow. Financial frictions close about half of this differential output response, through opposite responses of financial intermediaries in the two economies. Under FRMs, financial intermediaries take losses on their fixed-rate portfolios and raise mortgage spreads as policy tightens, deepening the contraction. Under VRMs, financial intermediaries are hedged against higher rates and compress their margins, softening the contraction.

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†Bank of Canada. E-mail: mirwin@bankofcanada.ca

‡Bank of Canada. E-mail: mvieyra@bankofcanada.ca

1 Introduction

Between 2021 and 2024 the central banks of the United States, Canada, the United Kingdom and Australia raised their policy rates by 4 to 5 percentage points (pp). Figure 1 shows what reached mortgage borrowers. In Australia, where most mortgages carry variable rates, the average rate on outstanding mortgages rose by 3.5 pp from its low. In Canada and the United Kingdom, where rates are fixed for two to five years, it rose by just under two. In the United States, where the typical mortgage is fixed for thirty years, it rose by 0.5 pp, even though the rate on new mortgages rose by more than 4 pp. The tightening reached American households slowly and only through new loans. For the holders of existing American mortgages, on the other hand, the tightening reached the fall in the value of their portfolio at once.

A fixed rate does not remove interest rate risk from a mortgage. It moves the risk from the borrower to the financial intermediary. Whether that transfer makes output less sensitive to the policy rate depends on what the financial intermediary does with the loss. If the loss is passed through to the financial intermediary's owners, the question is whether those owners cut spending by less than the borrowers whose payments did not rise. If instead the financial intermediary responds by lending less and charging more, part of the insurance that borrowers received is taken back through the supply of new mortgages. This paper asks how much of the difference between fixed and variable rates survives once we account for the supply side of the mortgage market.

Our main finding is that a substantial part of the stabilizing effect of fixed-rate mortgages disappears once the supply side is taken into account. When mortgage supply is perfectly elastic, so that intermediaries keep lending at a constant spread and pass their losses to savers, variable rates make the three-year output loss from a monetary tightening 60% larger than fixed rates do. Once intermediaries respond with elasticities estimated from mortgage market data, they close 41% of that gap over three years and almost two thirds of it on impact.

We obtain these results in a heterogeneous-agent New Keynesian model with housing. Households choose whether to rent or own, borrow against their house with long-term mortgages, and hold liquid savings. The contract rate on a mortgage resets with a fixed probability each quarter, and that probability is the only difference between our two economies: it is one in the variable-rate economy and 2% in the fixed-rate economy, so that an outstanding loan reprices about once every four years once sales and deaths are counted. Financial intermediaries hold the mortgages at market value and fund them with deposits and their own net worth. Their supply of mortgages follows the leverage equation of Chiang and Zoch (2026): leverage responds to the expected return on mortgages and to the policy rate with two lever-

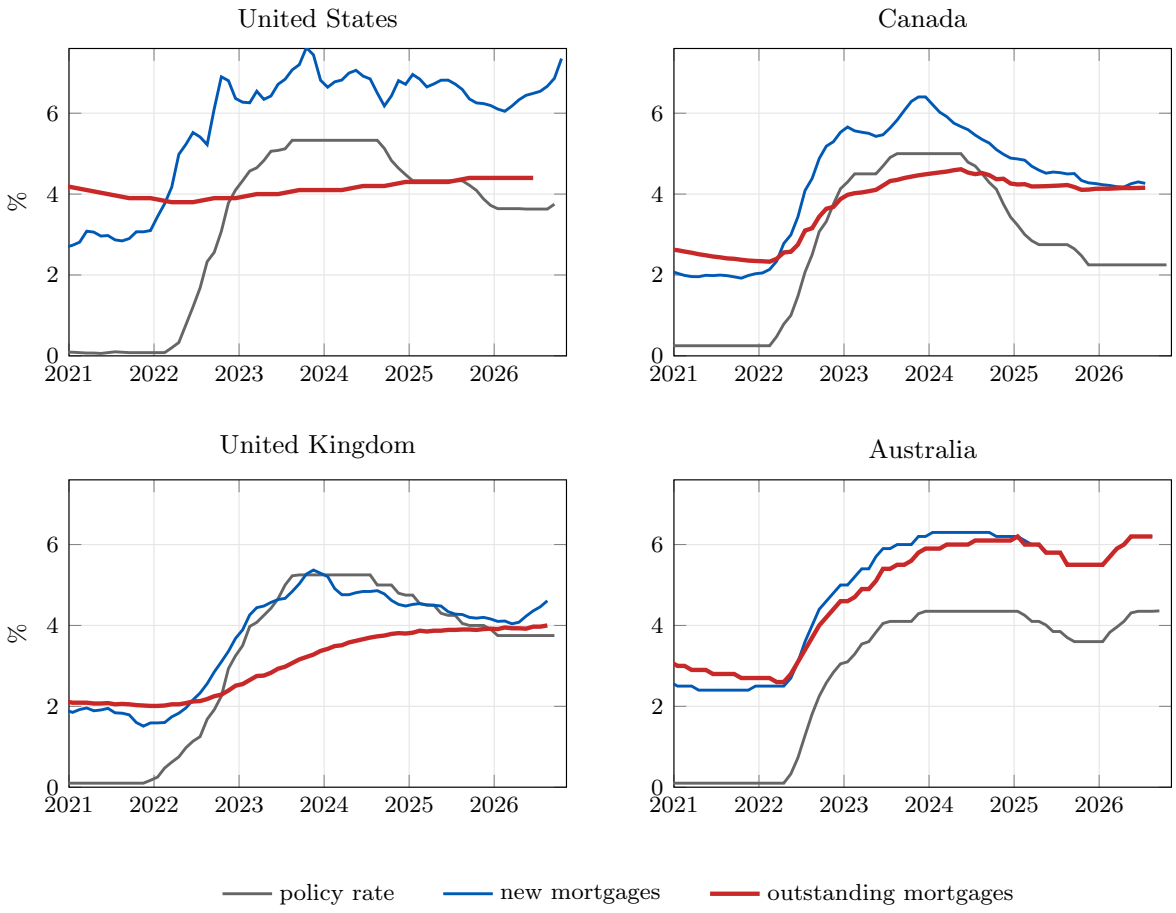


Figure 1: Policy rates and mortgage rates, 2021 to 2026

Notes: Monthly averages, in %. The policy rate is the effective federal funds rate, the Bank of Canada target rate, Bank Rate, and the RBA cash rate target. The rate on new mortgages is the Freddie Mac 30-year rate for the United States, the rate on funds advanced for residential mortgages (insured and uninsured, weighted by dollars advanced) from Statistics Canada, the Bank of England effective rate on new secured lending to households, and the RBA rate on new owner-occupier loans funded in the month. The rate on outstanding mortgages is the average rate in the FHFA National Mortgage Database (quarterly), the rate on outstanding residential mortgage balances from Statistics Canada, the Bank of England effective rate on outstanding secured lending, and the RBA rate on outstanding owner-occupier loans. Appendix D gives sources and construction.

age sensitivities, and net worth accumulates financial intermediaries' gains and losses. The equation nests the usual intermediary frictions, and it lets us discipline financial intermediaries with a handful of estimated responses instead of a particular friction. A mutual fund holds deposits, financial intermediary equity and government debt, so whatever financial intermediaries lose on their portfolio reaches savers through the return on their savings. Prices and wages are sticky, and a Taylor rule sets the policy rate.

We first shut the supply response down. Mortgage supply is then perfectly elastic at a constant spread, and the fall in the value of the mortgage portfolio is paid by savers through

the fund. In this economy a monetary tightening of 100 basis points (bps) reduces output 60% more over three years under variable rates than under fixed rates, because higher mortgage interest costs reduce borrowers' cash flow. Mortgage payments barely move in the fixed-rate economy, while the portfolio loses about 0.6% of its value on impact and the return on savings falls by more than the policy rate rises. Savers have low marginal propensities to consume, so the loss costs less output than the payments it replaces. So far the comparison is the one that most of the literature on mortgage contracts and monetary policy has studied, in which mortgage rates follow the policy rate and the balance sheet of the financial intermediary plays no role.[1] The rest of the paper asks what changes when it does.

To let financial intermediaries respond we need the sensitivities of their supply. We estimate the two leverage sensitivities and the response of net worth to financial intermediaries' returns by minimum distance, separately for each economy, on identified moments from the market it stands for. For the fixed-rate economy, where mortgages are funded mainly through mortgage-backed securities, we use three moments from the United States: the elasticity of demand for these securities with respect to their spread (Drechsler, Savov, Schnabl and Supera, 2024), the response of the mortgage spread to a monetary surprise (Gertler and Karadi, 2015), and the fall in financial intermediary equity after a rate surprise (English, Van den Heuvel and Zakrajšek, 2018). For the variable-rate economy, where mortgages are held on bank balance sheets, we use the same supply elasticity, the average pass-through of policy rates to bank lending rates within a quarter across studies (Gregor, Melecký and Melecký, 2021), and the pass-through to Swedish variable mortgage rates over two years (Kaplan and Njie, 2024). The estimates imply that the spread financial intermediaries require moves with the policy rate itself, and that the sign differs between the two economies. In the US estimates the spread rises by 0.86 bps for every bp of policy rate; in the variable-rate estimates it falls by 0.08. Neither set of estimates fits the other economy. These two responses match what happened in the 2021–2024 tightening, which we do not target: the spread of US mortgage rates over Treasury yields rose, while lenders in Australia, Canada and the United Kingdom compressed their margins on new loans. The difference lines up with how mortgages are funded. In the United States most are funded through mortgage-backed securities, whose investors demand more as yields rise; in variable-rate markets they

[1] Garriga, Kydland and Šustek (2017) and Kinnerud (2025) compare fixed-rate and adjustable-rate mortgages in models with long-term contracts, in which the adjustable rate is the short rate plus a constant spread and the fixed rate is the matching bond yield or, in Kinnerud (2025), a long rate that moves by an assumed share of the shock. Calza, Monacelli and Stracca (2013) make the comparison with one-period debt whose rate is either the policy rate or an average of expected policy rates. In none of them does the holder of the mortgages have a balance sheet of its own. Greenwald (2018) makes the same comparison in an appendix to a model of prepayable fixed-rate mortgages with a payment-to-income limit, and finds the larger response under fixed rates, because a fall in long rates sets off more prepayment.

are held by banks, whose deposit costs lag the policy rate.

With these estimates, financial frictions close about half of the gap between the two economies after a monetary tightening: 41% of the three-year output loss and 63% of the loss on impact. They do so through opposite responses of financial intermediaries in the two economies. Under variable rates, financial intermediaries are hedged against higher rates: their portfolio maintains its value, lending shrinks, and their margins compress, which softens the contraction by 7%. Under fixed rates, financial intermediaries take losses on their portfolios, their net worth falls by more than 1% on impact, and they raise the spread on new loans as the policy rate rises, which deepens the contraction by 13%. However, the losses themselves are not what moves the spread at the estimated sensitivities, contrary to what the usual balance-sheet accelerator would suggest. Although fixed-rate intermediaries lose on their portfolios, the losses barely change their leverage and add less than a basis point to the spread. What moves the spread is its response to the policy rate. This result fits with a market-based system like the American one. Mortgages there are priced by investors in mortgage-backed securities, whose demand is elastic enough that losses on existing holdings barely move the price of new credit, but who require a higher return when the policy rate rises.

Two further results add to the comparison between FRMs and VRMs: where the gap comes from, and how it depends on the shock. First, more than half of the gap after a tightening in the frictionless economy comes from the rate at which financial intermediaries qualify new borrowers, not from payments on existing loans. Buyer's are subject to a payment-to-income constraint when originating a loan: the payment at the current mortgage rate cannot exceed a share of income, as in Greenwald (2018), and in the variable-rate economy that rate rises one for one with the policy rate. Holding the qualifying rate at its steady-state value removes between a fifth and a quarter of that economy's output loss, and 54% of the gap. Second, the sign of the frictionless comparison depends on the shock. Fixed rates dampen the contraction after a fall in productivity, when the policy rate rises, and amplify the contraction after a fall in demand, when the policy rate falls and the easing reaches fixed-rate borrowers only through new and reset loans. In both cases the fixed-rate spread follows the policy rate, as it does after a tightening, and the supply response closes 71% and 86% of the gap, respectively, more than after the monetary shock. The rate on new fixed-rate loans averages the policy rate over several years, and these two shocks move the policy rate for longer.

Finally, the financial accelerator mechanism is stronger when supply is less elastic. If we double steady-state leverage to eight and make supply four times less elastic, the losses raise the spread on new fixed-rate loans substantially. Three quarters of the three-year gap then

closes, and on impact the fixed-rate economy contracts slightly more than the variable-rate one.

Related literature. The paper belongs to four strands of work. The first compares fixed and variable rates. Several papers find that monetary policy has larger effects where variable rates prevail, on consumption and house prices (Calza et al., 2013; Corsetti, Duarte and Mann, 2022; Hedlund, Larkin, Mitman and Ozkan, 2025) and on output (De Stefani and Mano, 2025). In contrast, Cloyne, Ferreira and Surico (2020) find that mortgagors raise their spending after a rate cut by about as much in the United States, where most mortgages carry fixed rates, as in the United Kingdom, where their payments fall nearly three times as much. They trace the response to the rise in household income rather than to lower payments. Models find larger effects under variable rates: Rubio (2011) and Calza et al. (2013) in New Keynesian models with collateral constraints, Garriga et al. (2017) with long-term nominal mortgages and flexible prices, Wong (2021) and Kinnerud (2025) in life-cycle models with refinancing, and Hedlund et al. (2025) for the divergence of house prices across countries during the 2021–2024 tightening. In these models the mortgage rate is fed in or follows the policy rate at a constant spread or an assumed pass-through, and Kinnerud (2025) shows that the gap between the two contracts depends on that pass-through. We add the other side of the contract: a financial intermediary whose portfolio is marked to market, whose losses reach savers, and whose supply of new mortgages responds to its balance sheet and to the policy rate at leverage sensitivities estimated separately for the two economies. The pass-through of the policy rate to mortgage rates is then an outcome of the model.

The second strand measures the mortgage channel of monetary policy at the household level. Di Maggio, Kermani, Keys, Piskorski, Ramcharan, Seru and Yao (2017) show that adjustable-rate borrowers whose payments fell at reset spent part of the savings and paid down debt, Flodén, Kilström, Sigurdsson and Vestman (2021) and Holm, Paul and Tischbirek (2021) that interest payments on variable-rate debt carry a tightening to the spending of Swedish and Norwegian households, and Cloyne et al. (2020) that mortgagors drive the aggregate consumption response in both the United States and the United Kingdom. When fixed-rate deals expire in the United Kingdom, most of the response of spending to a lower rate comes from borrowing against higher house prices rather than from lower payments (Foulis, Hazell, Mian and Tracey, 2026). Under long fixed rates the channel runs through refinancing (Beraja, Fuster, Hurst and Vavra, 2019; Berger, Milbradt, Tourre and Vavra, 2021; Eichenbaum, Rebelo and Wong, 2022). Greenwald (2018) shows that a payment-to-income limit makes new credit respond strongly to rates, a channel that works through new loans rather than through payments on existing ones. Our decomposition measures this

channel in the comparison between contracts: more than half of the frictionless gap after a tightening comes from the rate at which financial intermediaries qualify new borrowers. Our model also generates a strong response of household consumption to mortgage payments, because the borrowers who make them are concentrated among wealthy hand-to-mouth households, whose quarterly marginal propensity to consume averages 0.38 (Section 3.3).

The third strand asks who bears the interest rate risk that a fixed rate takes off the borrower, and what that does to the supply of credit. Fuster and Vickery (2015) show that the supply of fixed-rate mortgages in the United States depends on securitization, because originators are reluctant to keep their interest rate and prepayment risk, and Hoffmann, Langfield, Pierobon and Vuillemey (2019) find that the exposure of euro area banks to rates follows the fixation conventions of their national mortgage markets. Bank equity falls when rates rise unexpectedly (English et al., 2018). Drechsler, Savov and Schnabl (2021) argue that the deposit franchise hedges the long-term fixed-rate assets of banks, while DeMarzo, Krishnamurthy and Nagel (2024) find that franchise value falls as rates rise and adds to the losses on securities. Jiang, Matvos, Piskorski and Seru (2024) measure the mark-to-market losses of US banks after the 2022 tightening. Banks whose income reprices faster than their costs cut lending less when rates rise (Gomez, Landier, Sraer and Thesmar, 2021), and so do banks with more adjustable-rate mortgages (Altunok, Arslan and Ongena, 2024). In the United Kingdom, banks and building societies with more fixed-rate loans in their portfolios shift new lending between fixation lengths as the slope of the yield curve moves (Rajan, Rodriguez-Tous and Salgado-Moreno, 2025).

On the modeling side, many models of financial intermediaries use the leverage constraint introduced by Gertler and Kiyotaki (2010). We take the leverage equation of Chiang and Zoch (2026), which nests this constraint and other frictions with a few sensitivities. Greenwald, Landvoigt and Van Nieuwerburgh (2021), Diamond and Landvoigt (2022) and Arslan, Guler and Kuruşçu (2024) put constrained financial intermediaries inside housing models. Greenwald et al. (2021) find that indexing mortgage payments to aggregate house prices moves risk onto intermediaries and makes the financial system more fragile, the house-price counterpart of the question we ask about rates. In the New Keynesian model of Diamond, Landvoigt and Sánchez Sánchez (2025), the valuation losses of a mortgage intermediary pass through to savers, as they do through our fund. Guren, Krishnamurthy and McQuade (2021), Campbell, Clara and Cocco (2021) and Garriga and Hedlund (2022) compare mortgage designs, fixed and adjustable rates among them, with competitive intermediaries whose balance sheets do not limit lending.

Three recent papers bring the financial intermediary's side of the contract into a quantitative model and are the closest to ours. Elenev and Liu (2025) find that financial stability

risk is U-shaped in the fixation length, in a model with household default and banks funded by sticky deposits that face an exogenous short rate. Abad, Bigio, Garcia-Villegas, Marbet and Nuño (2026) show that fixed-rate and variable-rate banking systems transmit monetary policy differently only when bank solvency is at stake, and in a calibration to the euro area they find that lending responds to the policy rate by a third more in the fixed-rate system. Arslan and Guler (2026) show that in a flexible-price economy an inflation shock that enriches fixed-rate borrowers depletes bank capital and causes a recession. We study instead the transmission of monetary and other shocks to output in a heterogeneous-agent New Keynesian model with real mortgages, so that the Fisher channel is absent, and we discipline the financial intermediary's response in each economy with moments from the market it stands for rather than derive it from a friction. At the estimates the losses matter little and the response of the spread to the policy rate does the work. In our high-leverage case the losses matter, and on impact the fixed-rate economy contracts more than the variable-rate one, the direction that Abad et al. (2026) and Arslan and Guler (2026) find.

The fourth strand is the heterogeneous-agent New Keynesian literature. A rate change moves aggregate consumption through the covariance between households' exposures to it and their marginal propensities to consume (Auclert, 2019), and the mortgage contract decides whether the exposure sits with borrowers, many of them wealthy hand-to-mouth households (Kaplan and Violante, 2014; Kaplan, Moll and Violante, 2018), or, through the fund, with savers. Our household block builds on the housing models of Kaplan, Mitman and Violante (2020) and Hedlund, Karahan, Mitman and Ozkan (2021), the second of which embeds long-term mortgages into a heterogeneous-agent model with nominal rigidities. We solve the model in sequence space (Boppart, Krusell and Mitman, 2018; Auclert, Bardóczy, Rognlie and Straub, 2021b) with household Jacobians computed by the recursion of Bhandari, Bourany, Evans and Golosov (2023), extended so that each borrower carries its own contract rate.

The paper proceeds as follows. Section 2 presents the model, with the mortgage contract first and a subsection on what moves the spread. Section 3 presents the calibration and model fit. Section 4 compares the two economies when financial intermediaries pass their losses to savers and their supply of mortgages does not respond. Section 5 estimates the supply response of financial intermediaries in each economy and reads the estimates. Section 6 repeats the comparison with the estimated response, and Section 7 concludes.

2 Model

The economy is a heterogeneous-agent New Keynesian model with housing. Households face uninsurable income risk, choose whether to rent or own, finance purchases with long-term mortgages, and save in a single liquid asset. Idiosyncratic risk and incomplete markets generate a distribution of households over wealth, housing and debt, as in Huggett (1993) and Aiyagari (1994), and the housing and mortgage features are the standard ones in this literature (Piazzesi and Schneider, 2016). The contract rate on a mortgage resets with a probability that we set to one in the variable-rate economy and close to zero in the fixed-rate economy, and nothing else differs between the two. The financial intermediaries that hold the mortgages value them at market prices and supply them according to a leverage equation of the form proposed by Chiang and Zoch (2026), so that the losses a tightening imposes on fixed-rate financial intermediaries can feed back into mortgage spreads. And households own the financial intermediaries through a mutual fund whose return carries those losses to savers. Sticky prices and wages and a Taylor rule close the model, which keeps the hand-to-mouth channel that Kaplan et al. (2018) show matters for monetary transmission.

Time is discrete and a period is a quarter. All quantities, debts and returns are real, so inflation does not revalue outstanding balances and the central bank sets a real rate.[2] We present the mortgage contract first, since both channels run through it, then households, financial intermediaries, the fund, and the rest of the economy.

2.1 Mortgage contracts

A mortgage is a long-term loan secured by a house. A buyer who originates a balance m' at date t pays a fixed cost κ_m and borrows at the new-loan rate r_t^{new}. Two limits apply at origination:

$$m' \leq \chi^{LTV} p_t^h, \tag{1}$$

$$(r_t^{\text{new}} + \chi_2)\, m' \leq \chi_1\, w_t \ell_t e. \tag{2}$$

The first caps the loan at a fraction χ^{LTV} of the value p_t^h of the house, whose size is normalized to 1. The second caps the payment the financial intermediary imputes to the loan, at the current new-loan rate plus an offset χ_2 for taxes and insurance, at a share χ_1 of the borrower's labor income $w_t \ell_t e$, as in Greenwald (2018). The second limit is where the new-loan rate

[2] Nominal mortgage balances would add a Fisher channel, through which unexpected inflation transfers wealth from financial intermediaries to borrowers. We abstract from it to focus on the comparison between the two economies around the contract rate.

decides who can buy: a rise in r_t^{new} tightens it for every prospective borrower, whatever the rate existing borrowers pay.

Once originated, a loan with balance m and contract rate r^c pays interest $r^c m$ and repays a fraction γ of the principal each period, so that the balance carried into the next period is $(1-\gamma)m$. The contract rate is the borrower's own state. At the end of each period it resets to the current new-loan rate with probability ω and stays where it is otherwise:

$$r_{t+1}^c = \begin{cases} r_t^{\text{new}} & \text{with probability } \omega, \\ r_t^c & \text{otherwise.} \end{cases} \tag{3}$$

With $\omega = 1$ every contract reprices each period, and the economy has variable-rate mortgages (VRM). With $\omega = 0.02$ a contract reprices through resets about once every fifty periods; counting sales and deaths, about 6% of outstanding balances reprice each period, and the economy has fixed-rate mortgages (FRM). A borrower who sells, and the estate of a borrower who dies, repay the balance at par together with the last interest payment.

In the fixed-rate economy the reset probability plays the role of refinancing: a reset moves the borrower to the current new-loan rate, as refinancing would. Resets are equally likely whether the new-loan rate is above or below the contract rate. There is no default.[3] The two economies therefore share the same households, financial intermediaries, steady state and shocks, and differ only in ω.

2.2 Households

Households follow a perpetual-youth life cycle: each period, a household survives with probability ξ, and dying households are replaced by newborns so that the population mass remains constant at 1. A newborn starts as a renter with no liquid savings, and draws a discount factor from $\{\beta^L, \beta^H\}$ with probability π^H on the high value. Households derive utility from consumption c and from housing, and disutility from hours worked ℓ. Houses come in one

[3] A refinancing decision would make resets more frequent when rates fall and rarer when they rise, but to first order the average contract rate would still respond as if loans reset with a constant probability. Berger et al. (2021) show this for borrowers who refinance at random times whenever the market rate is below their own, a friction that fits US data much better than a fixed refinancing cost. The probability is the rate of moves plus the arrival rate of refinancing opportunities times the share of loans whose rate is above the market rate, as in their equation (6). The borrowers whom a small change in rates brings into or out of refinancing have rates close to the market rate, so the extensive margin plays no role, and rises and falls in rates of up to 100 bps have effects of nearly the same size. Our reset probability stands for the second term, and sales and deaths for the first. A constant probability leaves out the dependence of that share on past rates, through which the history of rates changes the response to policy (Berger et al., 2021; Eichenbaum et al., 2022). Default would add credit losses to the financial intermediary's balance sheet. We leave it out so that financial intermediaries bear only interest rate risk.

size, normalized to one, and owning one yields utility η. The period utility is therefore $u(c) + \eta - v(\ell)$ for an owner and $u(c) - v(\ell)$ for a renter, with u and v specified in Section 3. Since the two discount-factor types solve the same problem, we omit the type from the notation.

A household supplies ℓ_t hours, which a labor union sets uniformly across households, at the wage w_t per efficiency unit. Her labor productivity e follows a persistent process,

$$\log e' = \rho_e \log e + \upsilon', \qquad \upsilon' \sim N(0, \sigma_\upsilon^2), \tag{4}$$

and newborns draw their productivity from its ergodic distribution. Labor income is taxed at the rate τ, and every household receives the lump-sum transfer T_t. Savings a earn the return r_t^a, which in equilibrium is the return of the mutual fund described in Section 2.4, and households cannot borrow in the liquid asset.

Housing tenure is a discrete choice with taste shocks. A household that does not own a house chooses between renting and buying, and one that owns chooses between keeping and selling. Each choice carries a type-1 extreme value shock with scale σ_ε, so that the choice probabilities are logistic in the difference of the two values; Appendix A gives the expressions. The shocks capture idiosyncratic reasons to move that the model does not track, and they smooth the response of purchases and sales to prices. A non-owner's value at the start of the period is

$$V_t^n(a, e) = \mathbb{E}_\varepsilon \left[\max \left\{ V_t^{rent}(a, e),\ V_t^{buy}(a, e) + \varepsilon \right\} \right], \tag{5}$$

and an owner's value is

$$V_t^h(a, m, r^c, e) = \mathbb{E}_\varepsilon \left[\max \left\{ V_t^{own}(a, m, r^c, e) + \varepsilon,\ V_t^{sell}(a, m, r^c, e) \right\} \right]. \tag{6}$$

An owner's state includes the balance m and the contract rate r^c. In the fixed-rate economy the contract rate differs across owners, and it is this state that makes payments on existing loans insensitive to the policy rate.

Renters. A renter pays rent p_t^r, chooses consumption and savings, and starts the next period as a non-owner:

$$\begin{aligned} V_t^{rent}(a, e) = \max_{c, a'} \ & u(c) - v(\ell_t) + \beta \xi \, \mathbb{E} \left[V_{t+1}^n(a', e') \mid e \right] \\ \text{s.t.} \quad & c + a' + p_t^r = (1 - \tau) w_t \ell_t e + (1 + r_t^a) a + T_t, \\ & a' \geq 0. \end{aligned} \tag{7}$$

Buyers. A buyer pays the price p_t^h and a proportional transaction cost κ_h, and finances part of the purchase with a mortgage subject to the LTV and PTI constraints of Section 2.1:

$$V_t^{buy}(a,e) = \max_{c,a',m'} \; u(c) + \eta - v(\ell_t) + \beta\xi\, \mathbb{E}\left[V_{t+1}^h(a', m', r_t^{\text{new}}, e') \mid e\right] \tag{8}$$

$$\text{s.t.} \quad c + a' + p_t^h(1+\kappa_h) = (1-\tau)w_t\ell_t e + (1+r_t^a)a + m' - \mathbb{1}_{\{m'>0\}}\kappa_m + T_t,$$

$$\text{constraints (1) and (2)},$$

$$a' \geq 0.$$

In words, the buyer pays for the house and the transaction fee out of income, savings and the new loan net of its origination cost, and starts the next period as an owner whose contract rate is the new-loan rate of the period of purchase.

Owners. An owner who keeps the house pays maintenance $\delta_h p_t^h$ and property taxes τ_h, services the mortgage, and starts the next period as an owner with the balance net of amortization and a contract rate that may have reset:

$$V_t^{own}(a,m,r^c,e) = \max_{c,a'} \; u(c) + \eta - v(\ell_t) + \beta\xi\, \mathbb{E}\left[V_{t+1}^h(a', (1-\gamma)m, r_{t+1}^c, e') \mid e\right] \tag{9}$$

$$\text{s.t.} \quad c + a' + \delta_h p_t^h + \tau_h + (\gamma + r^c)\, m = (1-\tau)w_t\ell_t e + (1+r_t^a)a + T_t,$$

$$a' \geq 0,$$

with r_{t+1}^c given by (3). The payment $(\gamma + r^c)m$ is the channel through which a change in rates reaches an existing borrower. In the variable-rate economy r^c is last period's new-loan rate for every owner; in the fixed-rate economy it is the rate at which the owner last originated or reset.

Sellers. A seller repays the balance with its last interest payment, receives the value of the house net of maintenance, taxes and the transaction cost, rents for the period, and starts the next period as a non-owner:

$$V_t^{sell}(a,m,r^c,e) = \max_{c,a'} \; u(c) - v(\ell_t) + \beta\xi\, \mathbb{E}\left[V_{t+1}^n(a', e') \mid e\right] \tag{10}$$

$$\text{s.t.} \quad c + a' + p_t^r + (1+r^c)m = (1-\tau)w_t\ell_t e + (1+r_t^a)a + (1-\delta_h-\kappa_h)\, p_t^h - \tau_h + T_t,$$

$$a' \geq 0.$$

The timing imposes at least one period between selling a house and buying another.

Let μ_t^h and μ_t^n be the distributions of owners and non-owners over their states. We study

the response of the economy to unanticipated aggregate shocks. After a shock, households have perfect foresight over the path of aggregate prices, and we solve for that path to first order in sequence space (Boppart et al., 2018; Auclert et al., 2021b); Section 2.6 describes the method.

2.3 Financial intermediaries

A representative financial intermediary (FI) holds the economy's mortgages. It funds them with deposits, which are non-state contingent one-period debt that pays the policy rate r_t, and with its net worth n_t. Households hold both claims through the mutual fund. The financial intermediary values its portfolio at market prices, so a rise in the return it requires on mortgages lowers the value of loans whose coupons are locked in and raises its leverage measured at market value. We describe the financial intermediary with the sufficient-statistics representation of Chiang and Zoch (2026). To first order, the financial intermediary is summarized by two objects: a leverage equation, which gives the leverage it is willing to hold as a function of the expected return on its assets and its funding cost, and a law of motion for net worth. A broad class of frictions maps into these two objects, among them asset diversion, costly state verification, costly leverage and collateral constraints, each corresponding to particular values of their parameters. The strength of the accelerator is therefore set by a few parameters that can be estimated rather than derived from one microfoundation, and Section 5 estimates them for each economy. We add to this representation the pricing of mortgages: competitive pricing of new loans, and the valuation of a portfolio whose coupons reset slowly.

The portfolio. At the end of period t the financial intermediary holds the balances of owners who keep their house, net of amortization, and the loans originated that period:

$$M'_t = (1-\gamma)\, M_t^{\text{keep}} + M_t^{\text{buy}}. \tag{11}$$

Owners who sell, and the estates of those who die, repay at par. Each loan carries its contract rate, and the portfolio's coupon is the balance-weighted average of contract rates, an object the household block produces.

The required return and the spread. Let r_t^d be the one-period return the financial intermediary requires, at t, to hold mortgages from t to $t+1$. It is the price that clears the mortgage market. We write

$$s_t \equiv r_t^d - r_{t+1} - \iota \tag{12}$$

for the spread, the excess of the required return over the policy rate beyond the steady-state premium ι. The premium is a pricing wedge with no resource cost, and the financial intermediary block governs only the deviations of the spread from zero.

Pricing new loans. Lending is competitive, so the loans originated at t, discounted along the path of required returns, are worth exactly the cash lent, M_t^{buy}. To first order this par condition sets the new-loan rate equal to a weighted average of current and future required returns,

$$r_t^{\text{new}} - \bar{r}^m = \sum_{k\geq 0} w_k \left(r_{t+k}^d - \bar{r}^m\right), \qquad \sum_{k\geq 0} w_k = 1, \tag{13}$$

where $\bar{r}^m = \bar{r} + \iota$ is the steady-state mortgage rate. The weight w_k is the discounted share of the period's new loans that still carries its original rate k periods later. Balances lose that rate when they reset, and they leave the portfolio when they amortize or when the owner sells or dies. In the variable-rate economy every contract resets each period, so $w = (1, 0, 0, \dots)$ and $r_t^{\text{new}} = r_t^d$. In the fixed-rate economy the weights spread over the life of the loan: $w_0 = 0.06$ and the mean horizon is 12.7 quarters at our calibration (Appendix A), so the new-loan rate behaves like a long rate. This is the expectations hypothesis applied to the financial intermediary's required return.[4]

Valuing existing loans. Let V_t be the value of the portfolio's future payments discounted along the same path of required returns, and $Q_t = V_t/M_t'$ its price per unit of principal. In steady state the portfolio trades at par, $\bar{Q} = 1$. When required returns rise above the coupons locked into existing contracts, Q_t falls below one and the financial intermediary carries the unrealized loss $(1 - Q_t)M_t'$. New loans are priced at par, so a gap between Q_t and one reflects only loans whose coupon was set earlier. In the variable-rate economy every coupon resets each period and $Q_t = 1$: interest rate risk is borne by borrowers, not by the financial intermediary. In the fixed-rate economy few coupons reset and Q_t absorbs most of a change in required returns. The same discount curve prices both margins, so the two move together: at our calibration, each bp on the new-loan rate in the fixed-rate economy comes with a fall of about 4 bps in Q_t. Appendix A derives V_t and the weights.

The return on the portfolio. The financial intermediary earns the return R_t^m, paid at t, on the portfolio it held at the end of $t-1$. The return consists of the cash flow $\mathcal{C}_t$ (coupons, amortization, and balances repaid at par by sellers and estates) plus the end-of-t value of

[4] Chiang and Zoch (2026) use the same assumption in reverse, backing out expected one-period returns from forward rates on corporate yield curves.

the portfolio net of the loans originated at t, which by the par condition are worth M_t^{buy}:

$$1 + R_t^m = \frac{\mathcal{C}_t + V_t - M_t^{\text{buy}}}{V_{t-1}}. \tag{14}$$

In the period a shock hits, the change in required returns is unexpected and the return carries the revaluation of the portfolio, $R_t^m - \bar{r}^m \approx Q_t - 1$. Afterwards the portfolio earns exactly its required return, $R_{t+1}^m = r_t^d$.

Balance sheet and leverage. The financial intermediary's balance sheet at market value is

$$Q_t M_t' = D_t + n_t, \tag{15}$$

where D_t are deposits, and its leverage is $\Theta_t = Q_t M_t'/n_t$. A friction limits how far the financial intermediary will lever its net worth. Following Chiang and Zoch (2026), we work directly with the implied leverage equation. To first order, the leverage the financial intermediary is willing to hold at the end of t depends on the expected return on its portfolio and on its funding cost,

$$\Theta_t - \bar{\Theta} = \Theta_{r^K} \left(R_{t+1}^m - \bar{R}^m \right) - \Theta_{r^B} \left(r_{t+1} - \bar{r} \right). \tag{16}$$

The financial intermediary holds more leverage when it expects a higher return on its assets, and less when its funding is more expensive. The leverage sensitivities Θ_{r^K} and Θ_{r^B}, the change in leverage per unit of return, are the sufficient statistics for the friction.[5] Since $R_{t+1}^m = r_t^d$, equation (16) is a supply curve for mortgages as a function of the required return.

Net worth. The return on the financial intermediary's net worth, paid at t, is

$$r_t^N = r_t + \Theta_{t-1} \left(R_t^m - r_t \right), \tag{17}$$

since each unit of net worth funds Θ_{t-1} units of the portfolio, which earn R_t^m, and $\Theta_{t-1} - 1$ units of deposits, which cost r_t. Net worth follows the law of motion

$$\hat{n}_t = G_{r^N} \left(r_t^N - \bar{r}^N \right) + G_n \, \hat{n}_{t-1}, \tag{18}$$

[5] The leverage equation in Chiang and Zoch (2026) has a further term $\gamma_\Theta(\Theta_{t+1} - \bar{\Theta})$ that makes leverage forward looking. A static leverage equation, $\gamma_\Theta = 0$, covers costly state verification, costly leverage and collateral constraints; asset diversion in the manner of Gertler and Kiyotaki (2010) gives $\gamma_\Theta > 0$. We use the static form throughout, and the estimates in Section 5 are based on it.

where $\hat{n}_t$ is the proportional deviation of net worth from its steady state. The parameter G_{r^N} is the share of the return on net worth that stays in net worth: it equals one under full retention, and it is lower when payouts or equity issuance respond to returns. The persistence $G_n < 1$ captures payouts, entry and exit. Because households hold the financial intermediary's equity and its deposits through the same fund, these adjustments only move wealth between equity and deposits within the fund. Savers receive the portfolio's full market return whatever the value of G_{r^N}; the parameter governs only how much a gain or loss changes the financial intermediary's capacity to lend.

Market clearing. The mortgage market clears when the market value of the portfolio households hold equals what the financial intermediary is willing to hold given its net worth,

$$Q_t M_t' = \Theta_t \, n_t. \tag{19}$$

The portfolio M_t' is chosen by households through (11) and net worth is predetermined by (18), so the required return r_t^d adjusts until the leverage equation (16) is consistent with market-value leverage, and the new-loan rate then follows from the par condition (13).

The accelerator does not affect the steady state: the portfolio trades at par and all mortgage rates coincide with $\bar{r}^m = \bar{r} + \iota$. Net worth is $\bar{n} = \bar{M}'/\bar{\Theta}$, and its return is $\bar{r}^N = \bar{r} + \bar{\Theta}\,\iota$. Steady-state leverage $\bar{\Theta}$ affects no steady-state price; it only determines how the fund's claim on the portfolio divides between deposits and financial intermediary equity.

2.3.1 What moves the spread

The results of Sections 4 to 6 rest on one equation. Using $R_{t+1}^m = r_t^d$ and the definition of the spread, the leverage equation (16) inverts to

$$s_t = \underbrace{\frac{1}{\Theta_{r^K}}\left(\Theta_t - \bar{\Theta}\right)}_{\text{leverage}} + \underbrace{\left(\frac{\Theta_{r^B}}{\Theta_{r^K}} - 1\right)\left(r_{t+1} - \bar{r}\right)}_{\text{policy-rate term}}. \tag{20}$$

The spread has two parts. The first is the accelerator in the usual sense: a loss on the portfolio lowers net worth by more than it lowers assets, leverage rises above $\bar{\Theta}$, and the financial intermediary requires a higher return to hold the higher leverage. Its size per unit of leverage is $1/\Theta_{r^K}$. The second part moves the spread with the policy rate itself, at unchanged leverage. When $\Theta_{r^B} > \Theta_{r^K}$ a rise in the policy rate lowers the leverage the financial intermediary will hold at a given spread, so the spread rises with the policy rate;

when $\Theta_{r^B} < \Theta_{r^K}$ it falls. We call the coefficient $\Theta_{r^B}/\Theta_{r^K} - 1$ the policy-rate coefficient of the spread. The sign and size of the two terms are what Section 5 estimates, and they differ between the two economies.

Who pays the spread differs between the two economies. In the variable-rate economy every contract resets to $r_t^{\text{new}} = r_t^d$, so borrowers pay the spread on their whole balance one period later. In the fixed-rate economy only new loans pay it, and only as the average of current and future spreads that the par weights in (13) produce; existing borrowers are unaffected, and the financial intermediary bears the revaluation instead.

The no-accelerator economy is the limit in which mortgage supply is perfectly elastic, $\Theta_{r^K}, \Theta_{r^B} \to \infty$. The spread is then pinned at zero, $r_t^d = r_{t+1} + \iota$, and the financial intermediary passes the policy rate plus a constant premium through to mortgage rates. Net worth still moves with the financial intermediary's gains and losses, but it does not feed back into prices. The loss on a fixed-rate portfolio does not disappear in this economy. It reaches savers through the fund.

2.4 The mutual fund

Households do not hold the economy's assets directly. They save in a single asset, shares in a passive fund that holds the financial intermediary's deposits and equity, government debt, and the debt of a non-financial fund that collects firm profits and transaction fees (described below). The fund's assets at the end of t are

$$A_t^{MF} = D_t + n_t + B_{t+1}^g + B_{t+1}^F = Q_t M_t' + B_{t+1}^g + B_{t+1}^F, \tag{21}$$

where the second equality uses the financial intermediary's balance sheet (15). Deposits and the two kinds of debt earn the policy rate, while the financial intermediary's net worth earns r_t^N, so a share of the fund pays the portfolio-weighted return

$$r_t^{MF} = r_t + \left(r_t^N - r_t\right) \frac{n_{t-1}}{A_{t-1}^{MF}}, \tag{22}$$

realized at t on the portfolio assembled at $t-1$. Substituting the return on net worth (17) and the clearing condition (19) at $t-1$, the same return reads

$$r_t^{MF} A_{t-1}^{MF} = r_t \left(A_{t-1}^{MF} - Q_{t-1} M_{t-1}'\right) + R_t^m \, Q_{t-1} M_{t-1}'. \tag{23}$$

Savers earn the portfolio's market return on the part of the fund that finances mortgages and the policy rate on the rest, because deposits are safe. The whole revaluation of the portfolio

therefore reaches savers in the period it occurs, whatever the value of G_{r^N}. This is where the question of who bears the loss is settled: fixed rates spare existing borrowers the rise in rates, and the fund hands the corresponding fall in the value of their loans to the households that own the financial intermediary.

The fund return is the return on savings in the household problem, $r_t^a = r_t^{MF}$, and the market for savings clears when households' savings absorb the fund,

$$\int a_t' \, d\mu_t = A_t^{MF}. \tag{24}$$

In steady state $\bar{r} < \bar{r}^{MF} < \bar{r}^N$: households earn slightly more than the deposit rate because they own the levered financial intermediary through the fund.

2.5 Production, wages, government and monetary policy

Housing. The stock of owner-occupied housing is fixed at H^s, and the house price clears the market between owners who keep their house and buyers,

$$H^s = H_t^{\text{keep}} + H_t^{\text{buy}}. \tag{25}$$

For rental housing we follow Hedlund et al. (2021): landlords convert final goods into rental space with a linear technology, so the rent p_t^r is pinned down by that technology and does not respond to house prices. We scale the rental technology by the same productivity as goods production, so that a productivity shock does not move the relative price of rent.

Firms and prices. Monopolistically competitive firms produce with labor alone, $Y_t(f) = z_t L_t(f)$, where z_t is aggregate productivity. They reset prices at random intervals. This gives the Phillips curve

$$\pi_t = \kappa_p \left(\frac{w_t}{z_t} - \frac{\epsilon_p - 1}{\epsilon_p} \right) + \beta \xi \, \pi_{t+1}, \tag{26}$$

where π_t is inflation, w_t/z_t is real marginal cost, and ϵ_p is the elasticity of substitution across goods.

Unions and wages. A competitive labor packer buys employment services from a continuum of unions and sells aggregate labor L_t to firms, with elasticity of substitution ϵ_w across union tasks. Each union sets its wage at random intervals and calls on its members to supply hours uniformly, as in Erceg, Henderson and Levin (2000) and Schmitt-Grohe and Uribe (2005). Following Auclert, Rognlie, Souchier and Straub (2021a), the union's objective gives

the wage Phillips curve

$$\pi_t^w = \kappa_w \left(\frac{v'(L_t)}{\frac{\epsilon_w - 1}{\epsilon_w} w_t \, u'(C_t)} - 1 \right) + \beta \xi \, \pi_{t+1}^w, \tag{27}$$

where π_t^w is wage inflation and C_t aggregate consumption. Appendix A states the packer's and the unions' problems.

The non-financial fund. Firm profits and the transaction fees on house purchases, house sales and mortgage originations accrue to a non-financial fund owned by households through the mutual fund. In steady state it pays its revenue out as dividends D_t^F. Along a transition it smooths dividends: it borrows B_{t+1}^F at the policy rate to cover the change in its revenue and repays a fraction ϕ^g of its debt each period, the same rule the government follows. The rule keeps changes in profits and fees from driving the results through dividends.

Government. The government collects income and property taxes and the estates of the deceased, whose houses are sold and mortgages repaid, and pays for government consumption G_t, interest on its debt, and transfers T_t^g. Its budget constraint is

$$G_t + (1 + r_t) B_t^g + T_t^g = B_{t+1}^g + \tau w_t L_t + \tau_h H^s + \mathcal{B}_t, \tag{28}$$

where $\mathcal{B}_t$ are the estates. Government consumption is constant. Along a transition the government finances changes in its balance with debt and adjusts transfers to repay a fraction ϕ^g of the deviation of debt from its steady state each period,

$$T_t^g = \bar{T}^g - \phi^g \left(B_t^g - \bar{B}^g \right), \tag{29}$$

with debt given by the budget constraint. Total transfers to households are $T_t = T_t^g + D_t^F$.

Monetary policy. The central bank sets at t the real policy rate paid at $t + 1$ according to the rule

$$r_{t+1} = \rho_r r_t + (1 - \rho_r) \left[\bar{r} + \phi_\pi \left(\pi_t - \bar{\pi} \right) \right] + \zeta_t, \tag{30}$$

where ρ_r smooths the rate, ϕ_π is the response to inflation, and ζ_t is a monetary policy shock that decays geometrically. The same rate is the return on deposits, the financial intermediary's funding cost, and the return on the safe part of the fund.

2.6 Equilibrium and solution

An equilibrium is a path of household policy and value functions, distributions $\{\mu_t^h, \mu_t^n\}$, prices $\{\pi_t, \pi_t^w, w_t, p_t^h, p_t^r\}$, the policy rate r_t, the financial intermediary's prices and returns r_t^d, r_t^{new}, Q_t, R_t^m and r_t^N, the fund return r_t^{MF}, and financial intermediary quantities $\{\Theta_t, n_t, D_t\}$ such that households solve (5) and (6); the distributions evolve according to the policy functions and the exogenous processes; firms and unions set prices and wages according to (26) and (27); the financial intermediary's prices and quantities satisfy (13) to (19); the fund's return is (22); the government, the non-financial fund and the central bank follow their rules; and the markets for housing, mortgages, savings and labor clear. The goods market then clears by Walras' law.

Six aggregate paths are unknown once the household block is given: inflation, the house price, hours, transfers, the required return and the fund return. Six conditions pin them down: the markets for savings, housing, mortgages and labor, the transfer rule, and the fund identity. We solve for the response to a one-time unanticipated shock under perfect foresight in sequence space (Auclert et al., 2021b). The household block's Jacobians come from the recursion of Bhandari et al. (2023), which differentiates the optimality conditions at the steady state while holding each state's discrete choices and binding constraints fixed. We extend the recursion so that each borrower carries its own contract rate: the response of the household block to a change in the new-loan rate is tracked through the distribution of balances over the dates at which their rates were set, without adding a rate grid. Appendix B contains further details.

3 Calibration

We calibrate the model to the United States in 2019, at a quarterly frequency, in three steps. A first group of parameters is set outside the model, from the literature or from direct empirical counterparts. A second group is chosen so that the steady state matches moments from the data. The parameters of the financial intermediary block that differ between the two economies are estimated in Section 5 from the responses of mortgage markets to monetary policy. We then compare the model with moments we did not target, in particular the distribution of marginal propensities to consume, since the cash-flow channel is the product of the MPC and the change in payments. The two economies share this calibration. They differ only in the reset probability and, from Section 5 on, in the financial intermediary elasticities.

Description	Parameter	Value	Description	Parameter	Value
Risk aversion	σ	2.00	Owner-occupied housing stock	H^s	0.65
Inverse Frisch elasticity	ν	3.00	Housing depreciation (annual)	δ_h	1.5%
Survival probability	ξ	$1-1/180$	Property tax (annual, of value)	τ_h	1.03%
Share of patient households	π^H	0.30	Loan-to-value limit	χ^{LTV}	0.95
Productivity persistence	ρ_e	0.966	PTI offset (annual)	χ_2	3.52%
Productivity innovation variance	σ_v^2	0.017	Origination cost ($2,000)	κ_m	0.13
Price Phillips curve slope	κ_p	0.030	Amortization (annual)	γ	4%
Wage Phillips curve slope	κ_w	0.017	Mortgage premium (annual)	ι	1.8%
Elasticities, goods / labor types	ϵ_p, ϵ_w	20, 5	Contract reset, FRM / VRM	ω	0.02 / 1
Taylor rule	ρ_r, ϕ_π	0.8, 1.5	Steady-state FI leverage	$\bar{\Theta}$	4
Income tax rate	τ	0.30	Net-worth persistence	G_n	0.912
Government consumption / GDP	G/Y	0.14			
Government debt / annual GDP	$B^g/4Y$	0.79			
Debt gap repaid each quarter	ϕ^g	0.50			

Table 1: Parameters set outside the model

3.1 Parameters set outside the model

Table 1 lists the parameters set outside the model.

Preferences and demographics. Households have separable preferences over consumption and hours,

$$u(c) - v(\ell) = \frac{c^{1-\sigma}}{1-\sigma} - \phi \frac{\ell^{1+\nu}}{1+\nu}, \tag{31}$$

to which owning a house adds η. Risk aversion is $\sigma = 2$ and the inverse Frisch elasticity $\nu = 3$, both standard values, and the disutility of hours ϕ is set so that hours equal 1 in steady state. The survival probability $\xi = 1 - 1/180$ gives an expected adult life of 45 years. A share $\pi^H = 0.30$ of newborns draw the high discount factor. We calibrate η below.

Earnings. The persistence and innovation variance of labor productivity in (4) are 0.966 and 0.017, the quarterly values that McKay, Nakamura and Steinsson (2016) derive from the annual process estimated by Flodén and Lindé (2001).

Housing. The depreciation rate δ_h is set so that 1.5% of the housing stock depreciates each year, as in Kaplan et al. (2020), and the property tax τ_h so that annual property taxes equal 1.03% of the steady-state value of a house. Houses come in a single size, normalized to one, so the fixed stock of owner-occupied housing equals the homeownership rate; we set $H^s = 0.65$, the US rate in 2019 (U.S. Census Bureau, 2019).

Mortgages. Balances amortize at $\gamma = 0.01$ per quarter, so about 4% of principal is repaid each year and the average repayment horizon is 25 years. Following Kaplan et al. (2020), the loan-to-value limit is 0.95, so buyers put down at least 5%, and the origination cost κ_m

is the equivalent of \$2,000, which covers application, appraisal and closing fees. The offset in the payment-to-income limit, χ_2, stands for property taxes and insurance and is 3.52% of the balance per year, as in Greenwald (2018). Contracts reset with probability one in the variable-rate economy and 0.02 in the fixed-rate economy; with sales and deaths, about 6% of fixed-rate balances reprice each quarter.

Financial intermediaries. Steady-state leverage is $\bar{\Theta} = 4$, the effective leverage of intermediaries in Chiang and Zoch (2026), and the persistence of net worth $G_n = 0.912$ is their monthly estimate of 0.970 at a quarterly frequency. Steady-state leverage affects no steady-state price; it only determines how the fund's claim on the mortgage portfolio divides between deposits and financial intermediary equity. The premium ι is 1.8% a year, the gap in 2019 between the BEA rate on outstanding mortgages and the three-month Treasury bill rate. The leverage sensitivities and the net-worth elasticity are estimated in Section 5.

Prices, wages and policy. The elasticity of substitution across goods is $\epsilon_p = 20$, which gives steady-state profits of 5% of output, and across labor types $\epsilon_w = 5$. The slopes of the price and wage Phillips curves are 0.030 and 0.017. The Taylor rule smooths the rate with $\rho_r = 0.8$ and responds to inflation with $\phi_\pi = 1.5$.

Government. The income tax rate is $\tau = 0.30$, government consumption is 14% of output, and government debt is 79% of annual output. Along a transition, transfers repay half of the deviation of debt from its steady state each quarter, $\phi^g = 0.5$.[6]

3.2 Parameters chosen to match the steady state

The remaining parameters are chosen jointly so that the steady state matches the moments in Table 2; Table 3 reports their values. Although the parameters are chosen jointly, each moment is most informative about one of them, which Table 2 lists next to it.

Discount factors. The model needs both a low return on savings and a large share of hand-to-mouth households, and one discount factor cannot deliver both. The patient discount factor β^H is chosen so that the real return on the fund is 2% a year, and the impatient one β^L so that 20% of households are hand-to-mouth, the share in the 2019 Survey of Consumer Finances (SCF). The patient discount factor exceeds one, but the effective discount factor

[6]This is faster than is typical in the literature. With fast repayment, transfers absorb most of the fiscal consequences of a shock within a few quarters, which keeps the comparison between the two economies from depending on the path of public debt.

Moment	Data	Model	Parameter
Real return on savings (annual)	2.0%	2.0%	β^H
Share of hand-to-mouth households (SCF 2019)	0.20	0.20	β^L
Owners' median house value / income (SCF 2019)	2.91	2.96	p^h, η
Rent / house value (annual; 0.4 of the BEA yield of 5.84%)	2.34%	2.34%	p^r
Owner turnover (annual)	0.10	0.10	κ_h
PTI-bound purchase borrowers	0.18	0.18	χ_1
Buying response to a price subsidy	0.032	0.031	σ_ε
Not targeted			
Average quarterly MPC	0.20	0.20	
LTV-bound purchase borrowers	0.74	0.77	

Table 2: Targeted and untargeted moments

Notes: The shares of PTI-bound and LTV-bound purchase borrowers are from Greenwald (2018); the response of purchases to a price subsidy is from Berger, Turner and Zwick (2020). The average MPC is the quarterly consumption response to a one-time transfer, averaged over the stationary distribution.

β^L	β^H	η	p^h	p^r	κ_h	χ_1	σ_ε	T
0.968	1.002	0.63	15.8	0.092	2.16%	0.218	0.080	0.239

Table 3: Calibrated parameters

Notes: χ_1 is a share of gross labor income. The house price and the quarterly rent are in units of steady-state quarterly output. The implied steady-state rates are 0.82% a year on deposits, 2.0% on the fund, and 2.65% on mortgages.

including survival, $\beta^H\xi$, is below one. Permanent differences in patience are consistent with the evidence in Ganong, Greig, Noel, Sullivan and Vavra (2024), who find that households that are normally illiquid have high MPCs even when they are temporarily liquid.

Housing. Given the fixed supply of owner-occupied housing, the house price p^h is set so that the median ratio of house value to income among owners matches the 2019 SCF, 2.91, and the utility of homeownership η is chosen so that the housing market clears at that price. The rent p^r, pinned down by the productivity of rental supply, is set to 40% of the imputed rent of an owner-occupied house, which is 5.84% of its value per year in the BEA housing accounts and the Financial Accounts, so that a renter pays 2.3% of the house price a year; the discount stands in for the smaller size of rental units. The transaction cost κ_h matches an annual turnover of 10% among owners (Kaplan et al., 2020).

Tenure choice and borrowing limits. The scale of the taste shocks governs how strongly purchases respond to the value of buying, so we choose it to match the response of home

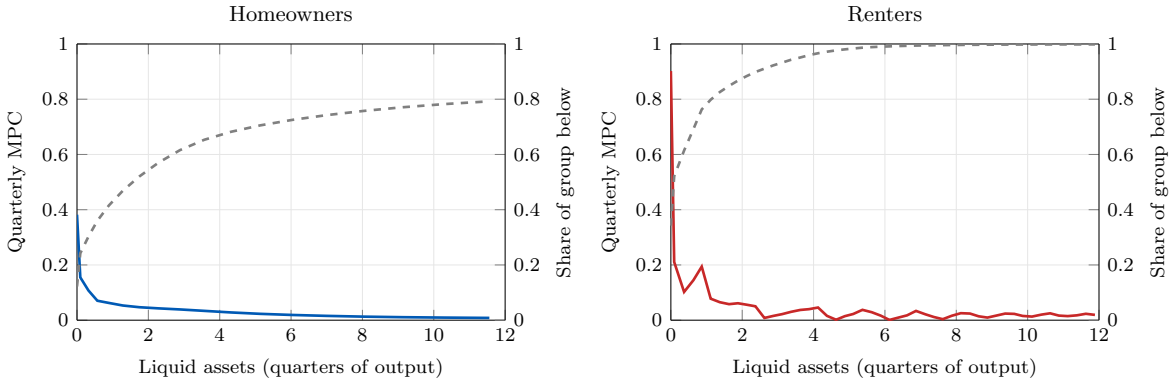


Figure 2: Marginal propensities to consume by liquid assets

Notes: Solid lines (left axis) show the quarterly MPC out of a one-time transfer of 1% of quarterly output, averaged over the stationary distribution of the other state variables. Dashed lines (right axis) show the cumulative share of each group with liquid assets below the value on the horizontal axis. The average MPC is 0.09 for homeowners and 0.39 for renters.

purchases to a price subsidy estimated by Berger et al. (2020). The buying and selling shocks share one scale, since both stand for idiosyncratic variation in the value of owning. The share of income available for mortgage payments, χ_1, matches the share of purchase borrowers bound by the payment-to-income limit, 18% in Greenwald (2018). Transfers T balance the government budget at the steady state.

The model also matches two moments we did not target. The average quarterly MPC is 0.20, in line with the evidence on consumption responses to transfers, and 77% of purchase borrowers are bound by the loan-to-value limit, against 74% in the data (Greenwald, 2018).

3.3 Consumption behavior

The share of hand-to-mouth households is targeted, but its split between poor and wealthy households is not, and neither is the distribution of MPCs. Both matter for our question. The payment channel of a rate change is the product of the change in payments and the MPC of the borrowers who pay it, and in the model those borrowers are concentrated among wealthy hand-to-mouth households: owners with a mortgage and little liquid wealth (Kaplan and Violante, 2014).

Figure 2 plots the quarterly MPC against liquid assets for homeowners and renters, together with the cumulative share of each group below each level of liquid assets. In both groups the MPC falls steeply with liquid assets and is highest at the borrowing constraint. The jump in the cumulative share at zero shows that many households sit at the constraint: 12% of homeowners and 35% of renters. They are constrained for different reasons. Constrained renters are mostly newborn or low-income households that cannot save out of their

Share of households (%)	Data	Model
Poor hand-to-mouth	8.2	12.1
Wealthy hand-to-mouth	11.6	7.9
Total	19.8	20.0

Table 4: Hand-to-mouth households

Notes: Data from the 2019 SCF, with the definition of Kaplan and Violante (2014). The total is a calibration target; its split is not.

labor income. Constrained homeowners are mostly recent buyers, who used their liquid savings for the down payment and carry large balances. Renters have a much higher average MPC than homeowners, 0.39 against 0.09, because more of them are constrained and their liquid wealth is lower. Constrained homeowners have an average MPC of 0.38, below the 0.90 of constrained renters and below the MPCs of wealthy hand-to-mouth households in Kaplan and Violante (2014) and Kaplan et al. (2018).

Table 4 compares the shares of poor and wealthy hand-to-mouth households with the 2019 SCF. In the data we follow Kaplan and Violante (2014): a household is hand-to-mouth if its liquid wealth is between zero and half of its income per pay period, taken to be two weeks, or if its liquid wealth is negative and within half of its pay-period income of its credit limit.[7] Hand-to-mouth households are wealthy if they hold positive illiquid wealth, such as housing or retirement accounts, and poor otherwise. In the model a household is hand-to-mouth if it sits at its borrowing constraint with no liquid savings; the wealthy ones are constrained homeowners, whose illiquid wealth is their home equity, and the poor ones constrained renters. The model generates a sizeable share of wealthy hand-to-mouth households, 7.9%, about two thirds of the 11.6% in the data. Because the total is targeted, it has more poor hand-to-mouth households than the data, 12.1% against 8.2.

4 Frictionless intermediation

This section compares the two economies when financial intermediaries do not respond. Mortgage supply is perfectly elastic at a constant spread, the fall in the value of the fixed-rate portfolio is paid by savers through the fund, and nothing feeds back from the financial intermediary's balance sheet into prices. This is the comparison that the literature on mortgage contracts and monetary policy has drawn, and it sets the benchmark that the accelerator is measured against in Section 6. We first describe the experiments, then the monetary tight-

[7] Liquid wealth includes cash, checking and savings accounts, directly held stocks and bonds, and mutual funds, net of credit card debt.

Output	Impact	Quarters 1–4	Quarters 1–12
VRM	−1.23	−5.34	−10.20
FRM	−0.80	−3.19	−6.36
Gap (FRM − VRM)	0.43	2.15	3.84
FRM loss relative to VRM	0.65	0.60	0.62

Table 5: Output after a monetary tightening without the accelerator

Notes: Output in % on impact and in pp-quarters for the sums. The last row is the ratio of the FRM response to the VRM response.

ening, then the two other shocks, and finally where the gap between the economies comes from.

4.1 The experiments

We study three shocks. The first is a monetary tightening: an innovation of 100 bps, annualized, to the Taylor rule (30), which decays with persistence 0.5. The second is a fall in productivity of 1% with persistence 0.79, after which the Taylor rule raises the policy rate. The third is a rise in all households' discount factors of 1% with persistence 0.72, a fall in demand after which the policy rate falls. Each shock hits four economies on the same steady state: the variable-rate and the fixed-rate economy, each with and without the accelerator. Without the accelerator the leverage sensitivities are infinite, so the spread is pinned at zero. The accelerator's effect on a variable is the response in the economy with it minus the response in the same economy without it.

Output, consumption, house prices, net worth and leverage are reported in % deviations from steady state, the portfolio price in % of par, and rates in annualized bps. Cumulative responses are sums of quarterly % deviations, in pp-quarters. The gap between the economies is the fixed-rate response minus the variable-rate response of output, so a positive gap means that the variable-rate economy contracts more.

4.2 A monetary tightening

Figure 3 shows the response of output, consumption, the house price, the policy rate, the rate borrowers pay on their outstanding loans, and the fund return to the tightening in the two economies. The variable-rate economy contracts more on every real margin. Output falls 1.23% on impact against 0.80%, consumption 1.6 against 1.0, and the house price about 2% against 1.3. Over three years the cumulative output loss is 10.2 in the variable-rate economy and 6.4 in the fixed-rate economy (Table 5): fixed rates cut the loss by 38%.

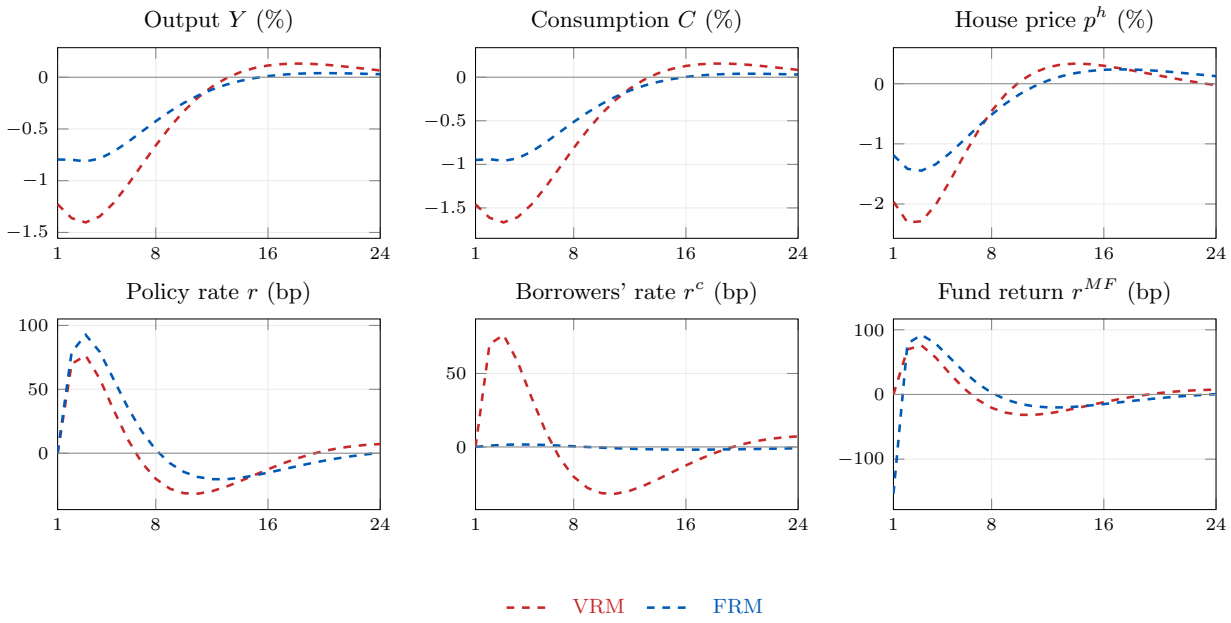


Figure 3: A monetary tightening without the accelerator

Notes: Responses to a 100 bp innovation to the Taylor rule with persistence 0.5, in the variable-rate (VRM) and fixed-rate (FRM) economies with mortgage supply perfectly elastic. Output, consumption and the house price in % deviations from steady state; rates in annualized bps. The borrowers' rate is the balance-weighted average contract rate on outstanding mortgages. The horizontal axis is in quarters.

The bottom row of the figure shows why. The policy rate rises by about 90 bps in the first quarter and falls back over two years, dipping below steady state in the second year as inflation undershoots. In the variable-rate economy the rate on outstanding loans follows it with a one-quarter lag, up 70 bps in the quarter after the shock. With outstanding mortgages equal to 5.8 quarters of output, that is extra interest of about 1% of quarterly output every quarter for the first year, paid by borrowers, who have the highest MPCs among owners. In the fixed-rate economy the rate on outstanding loans barely moves. Only new and reset loans carry the new rate, and the new-loan rate itself rises by only 15 bps, since it is an average of required returns over the 12.7-quarter horizon of the par weights in (13) and the tightening is expected to fade.

The loss does not disappear in the fixed-rate economy. The same discount curve that raises the new-loan rate lowers the value of existing loans. The portfolio price Q falls by 0.6% on impact, about 4 bps of value for every bp on the new-loan rate (Table A.1 in Appendix A), and the fund passes the loss to savers: the fund return falls by 153 bps on impact, in the quarter in which the policy rate rises by 90. In the variable-rate economy the fund return rises with the policy rate, because the portfolio holds its value and now pays more. The savers who take the loss hold most of the fund and have low MPCs, so the loss of about 3.4%

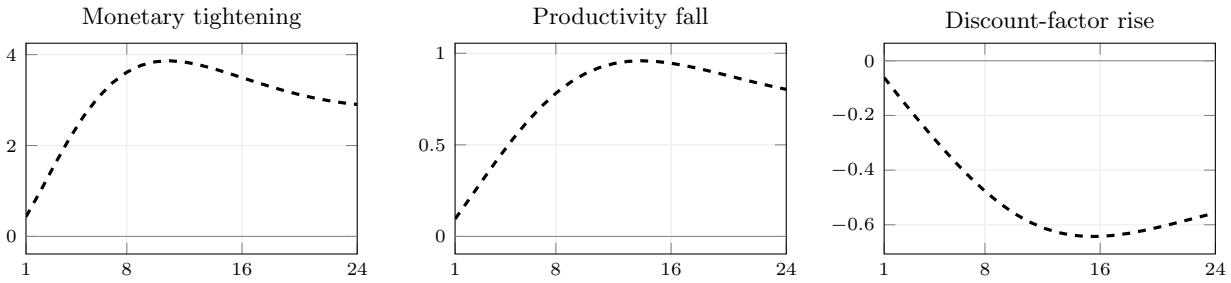


Figure 4: Cumulative output gap between the economies, three shocks, without the accelerator

Notes: Cumulative FRM minus VRM output response, in pp-quarters, after a 100 bp monetary tightening, a 1% fall in productivity, and a 1% rise in the discount factor. A positive gap means that the variable-rate economy contracts more. The horizontal axis is in quarters.

	Three-year output loss			
Shock	FRM	VRM	Gap	Policy rate
Monetary tightening	−6.36	−10.20	3.84	rises
Productivity fall	−1.01	−1.95	0.94	rises
Discount-factor rise	−3.54	−2.93	−0.61	falls

Table 6: Three shocks without the accelerator

Notes: Cumulative output responses over twelve quarters, in pp-quarters. The gap is the FRM minus the VRM response. The last column gives the direction of the policy rate under the Taylor rule.

of quarterly output in wealth costs less consumption than the payments it replaces. This is the fixed-rate economy's advantage without the accelerator: a transfer of the interest rate risk from borrowers with high MPCs to savers with low ones.

4.3 Supply and demand shocks

The ordering of the two economies after a tightening is not a property of fixed rates in general. It depends on the direction in which the policy rate moves relative to output. Figure 4 plots the cumulative gap between the two economies for the three shocks, and Table 6 reports the three-year sums.

After the fall in productivity the two economies line up as they do after the tightening. The shock raises inflation, the Taylor rule raises the policy rate, variable-rate borrowers pay more, and the variable-rate economy loses about twice as much output over three years, 1.95 against 1.01. After the rise in the discount factor the order reverses. Households want to save more, consumption and inflation fall, and the policy rate falls with them. Variable-rate borrowers get the easing on their whole balance within a quarter, which cushions their

		PTI at r_t^{new} (baseline)			PTI at $\bar{r}^m$			Due to PTI
	Δr_1^{new}	Impact	q1–4	q1–12	Impact	q1–4	q1–12	q1–12
VRM, no FA	69.9	−1.23	−5.34	−10.20	−1.10	−4.28	−7.74	24%
VRM, FA (VRM fit)	64.6	−1.14	−4.96	−9.46	−1.04	−4.08	−7.43	21%
FRM, no FA	14.9	−0.80	−3.19	−6.36	−0.75	−2.95	−5.97	6%
FRM, FA (FRM fit)	27.6	−0.99	−3.90	−7.20	−0.93	−3.54	−6.64	8%
Gap, no FA		0.43	2.15	3.84	0.34	1.33	1.77	54%
Gap, FA		0.16	1.06	2.26	0.12	0.53	0.80	65%

Table 7: Output with and without the qualifying-rate channel

Notes: Output in % on impact and in pp-quarters for the sums; the impact change in the new-loan rate in annualized bps. "Due to PTI" is the share of the three-year response, or of the gap, that disappears when financial intermediaries qualify buyers at the steady-state rate. FA: the economy-specific accelerator of Section 5.

consumption. Fixed-rate borrowers get it only on new loans, and the gain in the value of the portfolio goes to savers, who were already saving more. The fixed-rate economy loses 3.54 against 2.93, a fifth more. The pattern is a general one. Fixed rates stabilize output when the policy rate and output move in opposite directions, and they destabilize it when the two move together, because the contract then keeps the stabilizing movement of the rate away from the households who would spend it. Appendix C shows the full responses to the two shocks.

4.4 Where the gap comes from

The account so far attributes the gap after a tightening to the payments of existing borrowers. Part of it comes instead from who can borrow. Financial intermediaries qualify a buyer on the payment at the current new-loan rate, through the limit (2). In the variable-rate economy that rate rises 70 bps on impact; in the fixed-rate economy it rises 15. To measure how much of the gap comes from the qualifying rate, we hold the rate at which financial intermediaries qualify buyers at its steady-state value, so that the limit reads $(\bar{r}^m + \chi_2)\, m' \leq \chi_1 w_t \ell_t e$, and leave everything else unchanged, including the rate borrowers pay. Figure 5 shows the responses, and Table 7 the output sums, in the economies with and without the accelerator.[8]

The qualifying rate accounts for between a fifth and a quarter of the variable-rate economy's three-year output loss, and for 6 to 8% of the fixed-rate economy's. Without the accelerator the three-year gap falls from 3.84 to 1.77 when the qualifying rate is held fixed, so the channel is 54% of the gap; with the accelerator it is 65%. The share is smaller on

[8]The figure uses the economies with the economy-specific accelerator of Section 6, which are the baseline for the rest of the paper; the table reports both cases.

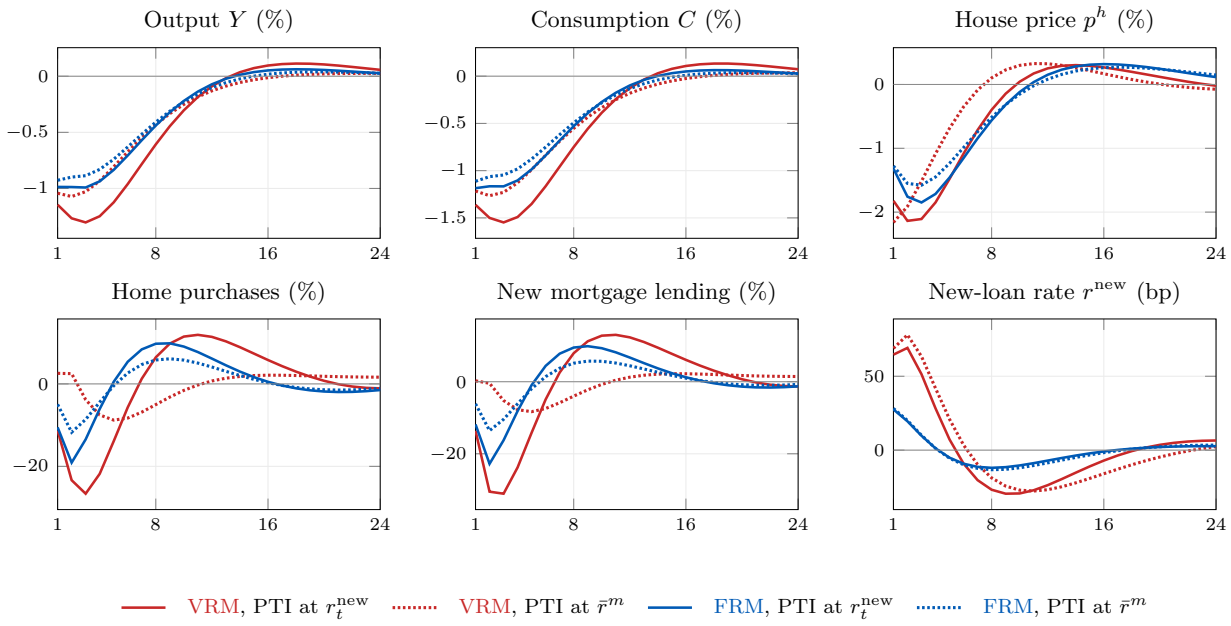


Figure 5: A monetary tightening with financial intermediaries qualifying buyers at the steady-state rate

Notes: Responses to the 100 bp tightening in the two economies with the accelerator at the economy-specific estimates of Section 5. Solid lines: the payment-to-income limit is applied at the current new-loan rate (baseline). Dotted lines: the limit is applied at the steady-state mortgage rate. Home purchases and new mortgage lending in % of their steady-state flows; the new-loan rate in annualized bps.

impact, about a quarter, and grows with the horizon, because the channel works through quantities that take time to reach output.

Those quantities are purchases. In the variable-rate economy home purchases fall 12% on impact at the baseline and rise 3% when the qualifying rate is fixed: the qualifying rate is the whole of the fall in purchases, and new mortgage lending falls 92% against 7 (Table C.1 in Appendix C). In the fixed-rate economy purchases fall 5% against 2, so the qualifying rate is about half of the fall. Purchases recover once rates fall below steady state in the second year and the limit loosens again, which is why the dotted and solid lines cross in the lower panels of Figure 5. The effect on consumption arrives in two steps (Table C.2). In the first year it is direct: households who cannot qualify stay renters, save for a larger down payment, and spend less. Over three years the direct part nets out and the general-equilibrium part carries the loss, through lower house prices and through lower hours and wages. Who can buy a house therefore matters about as much for the gap as what existing borrowers pay, and the two channels work through different households: the payment channel through owners with little liquid wealth, the qualifying-rate channel through renters at the margin of buying.

5 Mortgage supply in the two economies

The comparison of Section 4 gives the balance sheet of the financial intermediary no role. To let its supply of mortgages respond we need its sensitivities: how much leverage it gives up when its required return falls short, how much when its funding cost rises, and how much of a loss stays in its net worth. The fixed-rate and variable-rate economies stand for different markets, funded in different ways, so we estimate these sensitivities separately for each, on evidence from its own market. This section describes the two markets, the moments, the estimates and their fit, and what the estimates imply about what moves the spread.

5.1 Two markets

In the United States the typical mortgage is fixed for thirty years. At the end of 2024, 82% of outstanding loans were 30-year fixed-rate mortgages, 14% were 15-year fixed-rate mortgages, and 4% carried adjustable rates (Federal Housing Finance Agency, 2026). About four fifths of outstanding loans are acquired by the government-sponsored enterprises or insured by federal programs, and are funded through agency mortgage-backed securities. The interest rate risk in these loans sits with the investors who hold the securities: banks, the Federal Reserve, funds and foreign holders. The financial intermediary in our fixed-rate economy stands for them.

In Canada, the United Kingdom and Australia the risk sits mostly with borrowers. Canadian rates are fixed for at most five years; variable-rate loans were 19% of outstanding balances in January 2021 and a third by late 2022, as borrowers moved to them before the tightening (Statistics Canada, table 10-10-0006-01). British rates are fixed for two or five years and then revert to a variable rate. Most Australian mortgages carry variable rates that financial intermediaries reset at will. Figure 1 showed the consequence: the average rate on outstanding loans followed the policy rate up within a year in Australia and within two years in Canada and the United Kingdom. In all three countries mortgages are funded on bank balance sheets, with deposits as the main source and securitization a minority. These banks are hedged on their mortgage assets, and whatever rate risk they carry comes from the mismatch between their deposit funding and their short-fixation loans. The financial intermediary in our variable-rate economy stands for them.

Figure 6 shows what happened to the two kinds of financial intermediary during the tightening. In the United States the spread of the 30-year mortgage rate over the 10-year Treasury yield rose from 1.7 pp at the start of 2021 to 3.0 in mid-2023, and was still 2.3 at the end of 2024. Financial intermediaries required more to hold mortgages as rates rose, over and above the rise in the matched Treasury yield. At the same time the unrealized

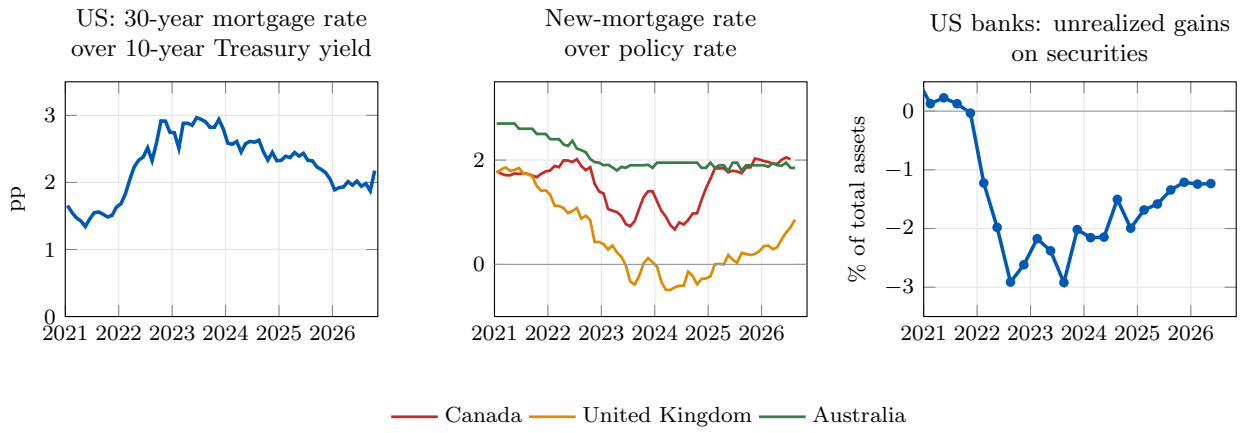


Figure 6: Financial intermediaries in the 2021 to 2024 tightening

Notes: Left: the Freddie Mac 30-year mortgage rate minus the 10-year Treasury yield, monthly averages. Middle: the rate on new mortgages (Figure 1; for Australia, new variable-rate owner-occupier loans) minus the policy rate, monthly. Right: unrealized gains and losses on investment securities of FDIC-insured institutions in % of their total assets, quarterly (Federal Deposit Insurance Corporation, 2026). Appendix D gives sources and construction.

losses on the securities held by US banks, most of them agency mortgage-backed securities and Treasuries, went from practically zero at the end of 2021 to 2.9% of total assets in the third quarter of 2022, about a quarter of the industry's equity at an assets-to-equity ratio of nine. In the three variable-rate and short-fixation markets the spread of the rate on new mortgages over the policy rate fell as the policy rate rose: by 0.9 pp in Australia between January 2021 and the end of 2023, by 0.4 in Canada, and by 1.7 in the United Kingdom.[9] Fixed-rate financial intermediaries lost on their portfolios and raised spreads; variable-rate financial intermediaries lost nothing on their portfolios and compressed their margins on new loans. The two financial intermediary blocks are estimated to reproduce these two patterns from identified moments, and no number in the figure is a target.

5.2 The moments

The parameters to estimate are the two leverage sensitivities in the leverage equation (16), Θ_{r^K} and Θ_{r^B}, and the share of returns that stays in net worth, G_{r^N}. Steady-state leverage $\bar{\Theta} = 4$ and the persistence of net worth $G_n = 0.912$ are fixed at the values of Section 3. Chiang and Zoch (2026) estimate the leverage equation from bank balance sheets and corporate bond returns. We instead estimate it from evidence on mortgage supply and its response to

[9] In Canada and the United Kingdom new loans are fixed for two to five years and priced off yields at those horizons, which were below the policy rate in 2023, so part of the fall is the term structure. Australian variable-rate loans are the cleanest case.

monetary policy, separately for each economy. Each moment is computed in the corresponding economy with the accelerator on, using yields constructed from the model's own path of policy rates under the expectations hypothesis. The model has no term premium, so its spreads reflect only the financial intermediary block and differences in maturity.

For the fixed-rate economy we target four moments from the US market. The first is the price elasticity of mortgage supply. Using Federal Reserve purchases of mortgage-backed securities as an instrument, Drechsler et al. (2024) find that the holdings of the marginal investors in these securities move by up to 20% of the stock per 100 bps of spread. The model counterpart is the change in the financial intermediary's desired holdings after a permanent 100 bp rise in the spread at given net worth, $\Theta_{r^K} \times 0.01/\bar{\Theta}$. The second and third are the response of the mortgage spread to a monetary surprise. Gertler and Karadi (2015) find that a surprise that raises the one-year yield by 20 bps raises the spread of the 30-year mortgage rate over the 10-year Treasury yield by 2 to 3 bps on impact and by 7 bps after two months. Averaging their monthly path by quarter, we target spread responses of 0.24 times the one-year yield response in the first quarter and 0.35 in the second; the model counterpart is the change in the new-loan rate net of the change in the model's ten-year yield, per unit of its one-year yield. The fourth is the response of financial intermediary equity. English et al. (2018) find that a 25 bp surprise rise in the level of the yield curve lowers bank stock values by 2.0 to 2.5%; we target the midpoint. The model counterpart is the impact change in net worth per 25 bps of the duration-matched yield, the average of the policy-rate path with the par weights of (13).

For the variable-rate economy we target three moments. The first is the price elasticity of supply, 20% per 100 bps, as in the US. No estimate exists for bank-funded variable-rate markets, so we treat the elasticity as a property of the intermediation technology that the two economies share. The second is the pass-through of the policy rate to lending rates within a quarter, 0.80, the average contemporaneous coefficient in the meta-analysis of Gregor et al. (2021). The third is the pass-through over two years, 0.93, the average pass-through of the policy rate to Swedish mortgage rates over policy cycles before 2022 (Kaplan and Njie, 2024). Variable-rate borrowers pay $r^c_{t+1} = r^d_t$, so the counterparts are the change in borrowers' rate in the quarter after the shock per unit of the policy rate, and the ratio of the two cumulative changes over quarters two to eight.

5.3 Estimates and fit

We choose the parameters to minimize the sum of squared relative deviations between the model and the data moments. Table 8 reports the estimates and Table 9 the fit.

Two features of the estimation deserve comment. First, the fixed-rate moments pin down

	Θ_{r^K}	Θ_{r^B}	G_{r^N}	Policy-rate coefficient $\Theta_{r^B}/\Theta_{r^K} - 1$
FRM (US moments)	320.6	594.7	0.31	+0.86
VRM (variable-rate moments)	327.2	299.6	0.31 (fixed)	−0.08

Table 8: Estimated financial intermediary parameters

Notes: Minimum-distance estimates on the moments of Section 5.2, with the static leverage equation (16). Steady-state leverage is 4 and the persistence of net worth 0.912 in both economies. In the no-accelerator economies of Section 4 both leverage sensitivities are 10^4.

Moment	Data	Own fit	Other fit
FRM economy: US moments			
MBS-holder price elasticity (% per 100 bp)	20	20.0	20.4
Mortgage spread, quarter 1 (× 1-year yield)	0.24	0.29	0.10
Mortgage spread, quarter 2 (× 1-year yield)	0.35	0.21	0.08
Intermediary equity on impact (% per 25 bp)	−2.0 to −2.5	-2.25	-1.15
Pass-through to the 30y FRM (not targeted)	0.20	0.35	0.17
VRM economy: variable-rate moments			
Price elasticity (% per 100 bp)	20	20.4	20.0
Pass-through, first quarter	0.80	0.90	1.82
Pass-through, quarters 2–8	0.93	0.76	1.49

Table 9: Fit of the two financial intermediary blocks

Notes: Own fit: the economy at its own estimates. Other fit: the fixed-rate economy at the variable-rate estimates, and the variable-rate economy at the fixed-rate estimates. The pass-through to the 30-year rate is not targeted; its data value is from McCormick and Ramaswamy (2026).

the long-run leverage sensitivity, but they do not separate the contemporaneous elasticity from the forward-looking term that Chiang and Zoch (2026) allow for. Along the ridge on which the long-run sensitivity equals its estimate the objective changes little, and its minimum is at a static leverage equation; minimizing over the forward-looking term in the variable-rate economy from several starting points also returns a static leverage equation. We therefore impose the static form throughout, and the long-run sensitivity is the object to compare with theirs: 320 here against 485 in their estimates. Second, the variable-rate moments carry no information about the dynamics of net worth. Variable-rate contracts reprice every quarter, the portfolio suffers no revaluation, and net worth barely moves, so none of the three moments involves G_{r^N}. We hold it at the fixed-rate estimate rather than at an arbitrary value.

Each block matches the elasticity of supply in its own economy. In the fixed-rate economy the model cannot produce the rising profile of the spread in Gertler and Karadi (2015): its spread peaks on impact and then decays, so the estimate splits the difference between the first- and second-quarter targets, 0.29 and 0.21 against 0.24 and 0.35. The equity loss is

matched. In the variable-rate economy the first-quarter pass-through is 0.90 against 0.80 and the two-year pass-through 0.76 against 0.93. Here too the model misses the timing, in the opposite direction: pass-through rises over the cycle in the data and declines in the model, because leverage keeps falling as lending shrinks and pulls the spread down. As an untargeted check, the pass-through of the policy rate to the 30-year rate in the fixed-rate economy is 0.35, against about 0.2 in reduced-form estimates; the model's long rate moves more than the data's because it has no term premium.

The last column of Table 9 shows that each set of estimates fits its own economy and misses the other's. Applied to the variable-rate economy, the fixed-rate estimates imply a pass-through of 1.82 in the first quarter and 1.49 over two years, well above one. Applied to the fixed-rate economy, the variable-rate estimates imply spread responses of 0.10 and 0.08 and an equity loss of 1.15%, half the data. This highlights the need to estimate separate sets of parameters for each economy.

5.4 What the estimates imply about the spread

The spread equation (20) has a leverage term and a policy-rate term, and the estimates settle their relative size. In both economies Θ_{r^K} is large, so changes in leverage move the spread little: a 3% fall in leverage, at steady-state leverage of 4, moves it by less than 4 bps. After the monetary tightening, the fixed-rate financial intermediary's net worth falls by 1.35% on impact, but its market-value leverage rises by only 0.11%, because the portfolio loses value along with net worth; the spread this generates is under 1 bp of the 67 bps observed (Table 11 in Section 6). The spread is governed instead by its response to the policy rate. In the fixed-rate economy $\Theta_{r^B}/\Theta_{r^K} = 1.86$: a rise in the policy rate lowers the leverage the financial intermediary will hold at a given spread by more than a rise in the required return raises it, so the spread rises by 0.86 times the change in the policy rate. In the variable-rate economy $\Theta_{r^B}/\Theta_{r^K} = 0.92$: the spread falls slightly with the policy rate and pass-through to borrowers is just below one.

We read the difference in the policy-rate coefficients as a reduced form for how mortgages are funded. Investors in mortgage-backed securities compare the mortgage yield with Treasuries of the same duration and require more when policy tightens, which is what the left panel of Figure 6 shows. Banks that fund variable-rate mortgages with deposits see their funding cost lag the policy rate and compete the spread on new loans down, which is what the middle panel shows. The model represents neither the deposit franchise nor the term premium, and the moments come from different countries, samples and methods, from high-frequency identification of US spreads to cycle-level pass-through in Sweden. The estimates should be read as the sign and size of two responses that the data support, not as

Output	Impact	Quarters 1–4	Quarters 1–12
VRM, no accelerator	−1.23	−5.34	−10.20
VRM, accelerator	−1.14	−4.96	−9.46
FRM, no accelerator	−0.80	−3.19	−6.36
FRM, accelerator	−0.99	−3.90	−7.20
Gap, no accelerator	0.43	2.15	3.84
Gap, accelerator	0.16	1.06	2.26
Share of the gap closed	63%	51%	41%

Table 10: Output after a monetary tightening with and without the accelerator

Notes: Output in % on impact and in pp-quarters for the sums. Each economy's accelerator is at its own estimates (Table 8). The gap is the FRM minus the VRM response.

the structure that produces them. On that reading, the financial intermediary's losses are a small part of what moves mortgage spreads after a tightening at the estimated sensitivities, and the response of its required return to the policy rate is the large part. Section 6 shows what that implies for the two economies.

6 When financial intermediaries cut supply

We now repeat the experiments of Section 4 with the accelerator on, each economy at its own estimates. The question is how much of the gap between the two economies survives once financial intermediaries respond, and through which of the two terms of the spread equation (20). We take the monetary tightening first, then the other two shocks, then a case in which financial intermediaries are more levered than the estimates imply.

6.1 A monetary tightening

Figure 7 shows the response of output, consumption, the house price, the spread, financial intermediary net worth and the new-loan rate to the tightening in the four economies, and Table 10 reports output. The accelerator closes 41% of the three-year gap between the economies, and 63% of the gap on impact. It does so from both sides. The fixed-rate contraction deepens: the three-year loss rises from 6.36 to 7.20, 13% more. The variable-rate contraction softens: the loss falls from 10.20 to 9.46, 7% less. Of the 1.58 pp-quarters closed over three years, 53% comes from the fixed-rate side and 47% from the variable-rate side. Fixed rates still stabilize output after a tightening, but by 24% over three years rather than 38.

The two financial intermediaries respond in opposite directions, as the spread panel of

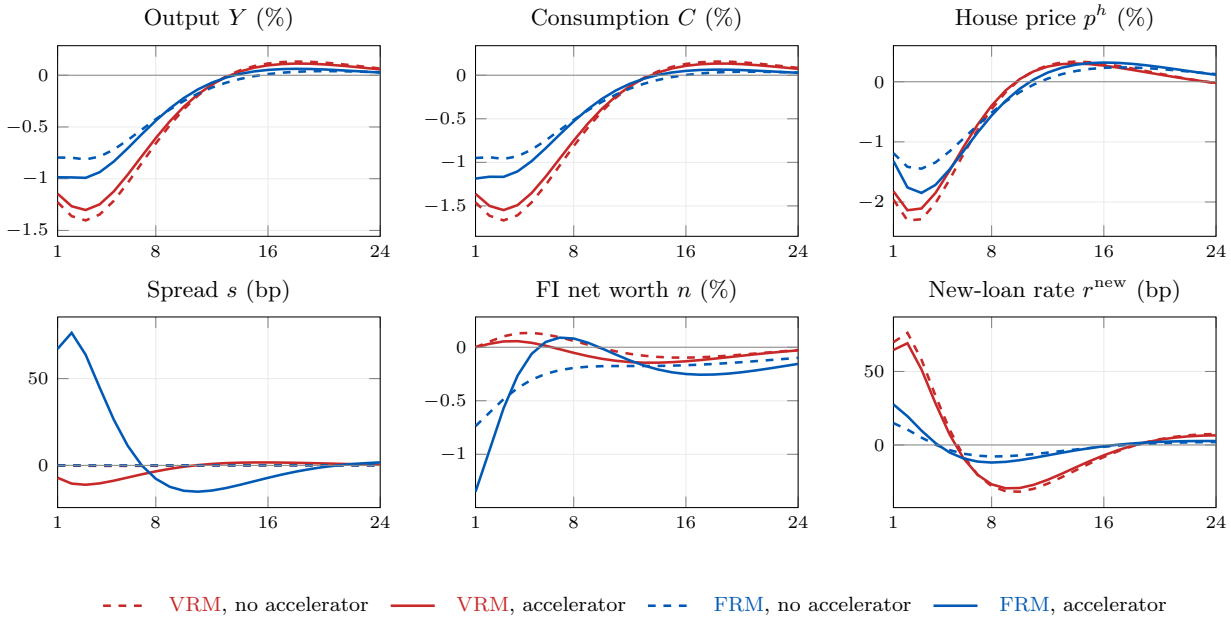


Figure 7: A monetary tightening with the economy-specific accelerator

Notes: Responses to the 100 bp tightening in the variable-rate and fixed-rate economies, each with the accelerator at its own estimates (solid) and without it (dashed). Output, consumption, the house price and net worth in % deviations from steady state; the spread and the new-loan rate in annualized bps. Figure C.1 in Appendix C shows leverage and the portfolio price.

Figure 7 and Table 11 show. The fixed-rate financial intermediary's spread rises by 67 bps on impact and 76 in the second quarter, and tracks the policy rate down thereafter. The new-loan rate rises by 13 bps more than without the accelerator on impact, which is small next to the spread because the par coupon averages the spread over the horizon of the weights and the spread is expected to fade. The fixed-rate financial intermediary also loses: its portfolio falls 1.1% in value, almost twice the fall without the accelerator because the higher new-loan rate discounts existing loans more, and its net worth falls 1.35%. But its market-value leverage barely moves, up 0.11% on impact and below steady state from the second quarter, because assets fall with net worth and lending contracts. At the estimated sensitivity the leverage term contributes under 1 bp to the spread. The variable-rate financial intermediary's spread falls by 7 to 11 bps for two years. Its portfolio holds its value and its net worth is flat; leverage falls as lending shrinks, and the small negative policy-rate coefficient takes the spread below zero. Borrowers in the variable-rate economy therefore pay 5 to 7 bps less on their whole balance than without the accelerator.

Two channels could close the gap: the losses that fixed rates impose on financial intermediaries, or the response of the spread to the policy rate. The paths above give the answer. Net worth falls by more than 1% in the fixed-rate economy, but the leverage it creates is

Variable-rate economy, variable-rate estimates							
t	1	2	3	4	6	8	12
$\hat{\Theta}_t$ (%)	−0.22	−0.77	−1.21	−1.45	−1.47	−1.11	−0.25
$\hat{n}_t$ (%)	−0.00	0.03	0.05	0.06	0.01	−0.05	−0.14
s_t (bp)	−7.1	−10.5	−11.2	−10.4	−7.0	−3.4	0.8
Δr_t^{new} (bp)	−5.3	−7.1	−6.7	−5.2	−1.6	1.2	2.6
Fixed-rate economy, US estimates							
t	1	2	3	4	6	8	12
$\hat{\Theta}_t$ (%)	0.11	−0.23	−0.42	−0.50	−0.41	−0.21	0.14
$\hat{n}_t$ (%)	−1.35	−0.96	−0.57	−0.26	0.05	0.08	−0.12
s_t (bp)	67.2	76.4	63.7	44.8	11.3	−7.7	−14.5
Δr_t^{new} (bp)	12.7	9.1	4.7	1.1	−3.2	−4.1	−1.8

Table 11: Financial intermediary variables after the monetary tightening

Notes: Leverage and net worth in % deviations from steady state, the spread in annualized bps, and the change in the new-loan rate relative to the same economy without the accelerator, in annualized bps. t is the quarter after the shock.

priced at a few bps, so the duration loss does not account for the accelerator's effect on output at the estimates. The response of the spread to the policy rate does. The spread tracks the policy rate at 0.86 in the fixed-rate economy and at a small negative coefficient in the variable-rate economy, and those two coefficients are the whole of the response.

Table 12 shows how the accelerator's effect on consumption reaches households, by decomposing it into the contributions of each input of the household block, holding the others at their no-accelerator paths. In the fixed-rate economy the first-quarter effect comes almost entirely through the fund return: the larger loss on the portfolio lowers savers' return, and savers cut consumption by 0.12% of its steady-state value on impact. Over three years that term nets out and the loss comes through general equilibrium, through the house price, hours and wages, with the new-loan rate itself contributing little, since few households borrow at it in any quarter. In the variable-rate economy the gain comes through the borrowing rate, which every owner pays, and through the house price, hours and wages that a milder contraction supports. The accelerator reaches fixed-rate households as owners of the financial intermediary and variable-rate households as its borrowers.

6.2 Supply and demand shocks

The same two policy-rate coefficients act on the other shocks, and they close more of the gap. Figure 8 plots the cumulative gap with and without the accelerator for the three shocks, and Table 13 reports the sums. After the fall in productivity the accelerator closes 71% of the gap: the fixed-rate loss rises by 51%, from 1.01 to 1.52, as the spread rises 10

	Fixed-rate economy			Variable-rate economy		
Input	Impact	q1–4	q1–12	Impact	q1–4	q1–12
House price p^h	0.01	-0.13	-0.67	0.02	0.11	0.38
New-loan rate r^{new}	-0.04	-0.21	0.15	0.03	0.16	0.15
Saving return r^{MF}	-0.12	-0.17	0.08	0.00	-0.01	-0.04
Hours N	-0.05	-0.17	-0.27	0.03	0.11	0.24
Wage w	-0.02	-0.08	-0.16	0.02	0.08	0.21
Transfers T	-0.01	-0.08	-0.12	0.00	0.00	-0.06
Total ΔC	-0.24	-0.84	-0.98	0.10	0.45	0.89

Table 12: The accelerator's effect on consumption, by input of the household block

Notes: The response of consumption in the economy with the accelerator minus the response without it, in % of steady-state consumption on impact and in pp-quarters for the sums, attributed to each input of the household block through the household Jacobians. The new-loan rate enters the fixed-rate economy through new and reset loans and the qualifying rate, and the variable-rate economy through every outstanding loan.

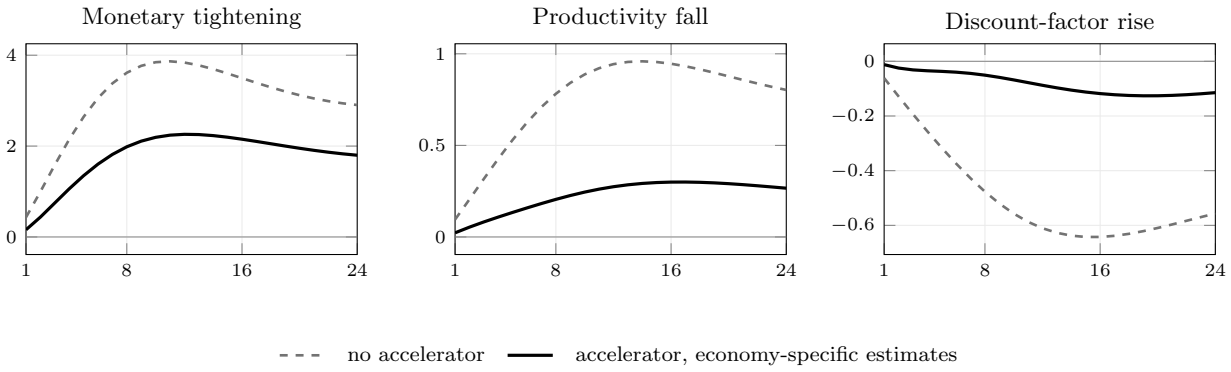


Figure 8: Cumulative output gap between the economies, three shocks, with and without the accelerator

Notes: Cumulative FRM minus VRM output response, in pp-quarters, after the 100 bp monetary tightening, the 1% fall in productivity, and the 1% rise in the discount factor, without the accelerator (dashed) and with each economy's accelerator at its own estimates (solid). A positive gap means that the variable-rate economy contracts more.

to 16 bps with the policy rate, and the variable-rate loss falls by 8%. After the rise in the discount factor it closes 86%, from the opposite directions. The policy rate falls, the fixed-rate financial intermediary's spread falls with it, the new-loan rate falls by more than without the accelerator, and the portfolio gains value; the fixed-rate loss falls by 12%, from 3.54 to 3.11, while the variable-rate loss rises by 3%. The fixed-rate spread follows the policy rate whichever way it moves, which strengthens transmission in both cases, and that is why it closes the gap in both.

The share closed rises from the monetary shock to the productivity shock to the demand

	Three-year loss, no accelerator		Three-year gap		Closed	Accelerator effect (%)	
Shock	FRM	VRM	no acc.	acc.	(%)	FRM	VRM
Monetary tightening	−6.36	−10.20	3.84	2.26	41	13.2	−7.3
Productivity fall	−1.01	−1.95	0.94	0.27	71	50.8	−8.0
Discount-factor rise	−3.54	−2.93	−0.61	−0.09	86	−12.2	3.1

Table 13: Three shocks with and without the accelerator

Notes: Cumulative output over twelve quarters in pp-quarters. The accelerator effect is the % change in the three-year loss of each economy when its accelerator is turned on. Figures C.3 and C.4 in Appendix C show the responses.

shock because the two sides of the comparison depend on different parts of the rate path. The gap without the accelerator is driven by what variable-rate borrowers pay in the first year, so it tracks the short end of the path. The accelerator's effect on the fixed-rate economy runs through the new-loan rate, which averages required returns over the 12.7 quarters of the par weights and prices the whole portfolio, so it tracks the long end. The ratio of the long end to the short end rises across the three shocks, from the monetary shock, which fades with persistence 0.5, to the two shocks whose rate paths last longer, and with it rises the part of the gap that the policy-rate term can close.

6.3 A high-leverage financial intermediary

The estimates describe financial intermediaries in normal times. To see what the accelerator does when financial intermediaries carry more leverage, we raise steady-state leverage from 4 to 8. We also make their supply of mortgages four times less elastic by setting both leverage sensitivities to a quarter of their estimates, while keeping each economy's policy-rate coefficient and net-worth elasticity. We call this the high-leverage case. Two ratios change. The spread per 1% change in leverage, $\bar{\Theta}/\Theta_{r^K}$, is eight times larger, and the response of net worth to the return on the portfolio, $G_{r^N}\bar{\Theta}$, doubles from 1.24 to 2.48. The price elasticity of supply falls from 20 to 2.5% per 100 bps, which is a crisis value rather than an estimate. Steady-state prices are unchanged; financial intermediary net worth halves and deposits rise. Figure 9 shows the responses to the monetary tightening and the cumulative gap, and Table 14 reports output.

With a high-leverage financial intermediary the losses matter. In the fixed-rate economy net worth falls 3.1% on impact and leverage rises 1.7%, and the less elastic mortgage supply prices that leverage at 66 bps, which with the policy-rate term takes the first-quarter spread to 134 bps. The spread then falls below its path at the estimates from the fourth quarter, as lending contracts and leverage falls. The fixed-rate contraction deepens by 30% on impact,

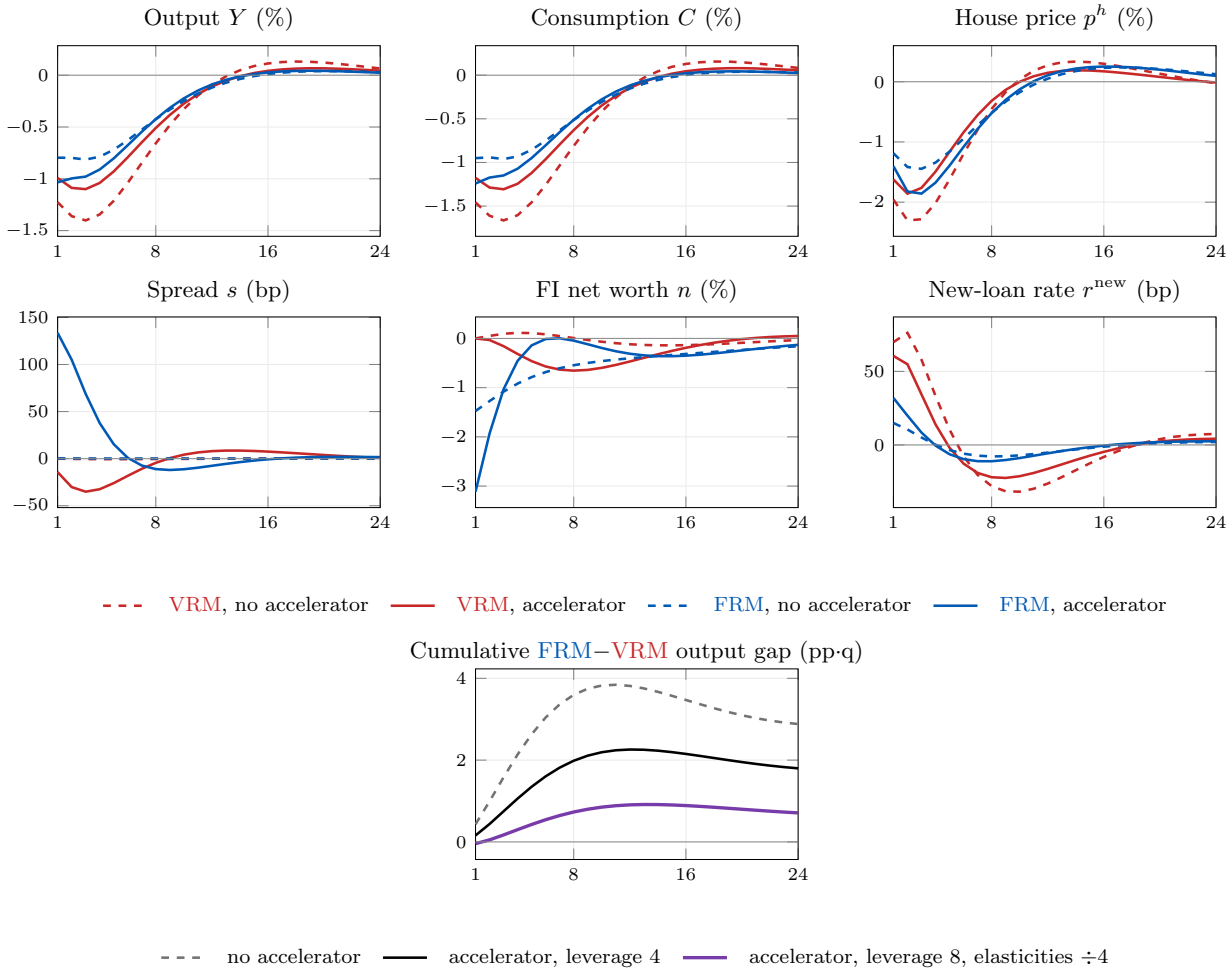


Figure 9: A monetary tightening with a high-leverage financial intermediary

Notes: Steady-state leverage 8 and leverage sensitivities at a quarter of the estimates, with each economy's policy-rate coefficient and net-worth elasticity unchanged. Top two rows as in Figure 7. Bottom: the cumulative FRM minus VRM output gap without the accelerator, with the accelerator at the estimates, and with the high-leverage financial intermediary. Figure C.2 in Appendix C shows leverage and the portfolio price.

against 24% at the estimates, but by 12% over three years, about the same as before. In the variable-rate economy the spread falls by 14 to 35 bps as lending shrinks and leverage falls with the less elastic supply, and the contraction softens by 21% over three years. Three quarters of the three-year gap closes, 73% of it from the variable-rate side, and on impact the fixed-rate economy contracts more than the variable-rate one. The lesson of the high-leverage case is that the sign of the comparison on impact is not fixed by the contract. An economy that insures its borrowers against interest rate risk can be the less stable of the two once the financial intermediaries who carry that risk are levered enough, and even at the estimates most of the insurance that fixed rates give borrowers is taken back through the

Output	Impact	Quarters 1–4	Quarters 1–12
VRM, no accelerator	−1.23	−5.33	−10.18
VRM, high-leverage financial intermediary	−0.99	−4.22	−8.05
FRM, no accelerator	−0.80	−3.19	−6.37
FRM, high-leverage financial intermediary	−1.03	−3.92	−7.14
Gap, no accelerator	0.43	2.14	3.81
Gap, high-leverage financial intermediary	−0.04	0.30	0.91
Share of the gap closed	110%	86%	76%
Share closed at the estimates	63%	51%	41%

Table 14: Output after a monetary tightening with a high-leverage financial intermediary

Notes: Output in % on impact and in pp-quarters for the sums. Steady-state leverage 8 and leverage sensitivities at a quarter of the estimates. The last row repeats Table 10.

supply of new loans.

7 Conclusion

A fixed-rate mortgage moves interest rate risk from the borrower to the financial intermediary. This paper asked what the financial intermediary does with it, and what that does to the comparison between fixed and variable rates. We built a heterogeneous-agent New Keynesian model with housing in which the contract rate resets with a given probability, financial intermediaries hold mortgages at market value and supply them according to a leverage equation, and a fund carries financial intermediaries' gains and losses to savers. We estimated the leverage equation separately for the US market, funded through mortgage-backed securities, and for bank-funded variable-rate markets.

When mortgage supply does not respond, a monetary tightening reduces output 60% more over three years under variable rates than under fixed rates, because higher interest costs reduce borrowers' cash flow while the fall in the value of fixed-rate loans reaches savers with low propensities to consume. More than half of that difference comes from the rate at which financial intermediaries qualify new borrowers rather than from payments on existing loans. When financial intermediaries respond, about half of the difference disappears, through opposite responses in the two economies: fixed-rate financial intermediaries take losses and raise spreads as the policy rate rises, and variable-rate financial intermediaries, hedged on their portfolios, compress their margins. At the estimated sensitivities it is the response of spreads to the policy rate, not the losses themselves, that does the work. When financial intermediaries are more levered the losses matter too, three quarters of the difference disappears, and on impact the fixed-rate economy contracts more than the variable-rate one. The same mechanism closes most of the gap after supply and demand shocks, where

the frictionless comparison itself changes sign with the direction of the policy rate.

The model leaves out three things that would change the numbers without, we think, changing the conclusion. Resets at a constant rate capture refinancing to first order, but not its dependence on the history of rates (Berger et al., 2021; Eichenbaum et al., 2022). Default would add credit losses to financial intermediaries' balance sheets in a contraction. And the financial intermediary blocks are reduced forms for two kinds of funding; a model of deposits and of the term premium would say where the estimated responses of spreads to the policy rate come from. The moments we estimate on also come from different countries and samples. In ongoing work we use Canadian bank regulatory data, in which fixed and variable loans sit on the same balance sheets, to estimate the financial intermediary's leverage sensitivities conditional on the fixation length of its portfolio.

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A Model details

A.1 Tenure choice

With type-1 extreme value shocks of scale σ_ε, the expected value of a non-owner at the start of the quarter and the probability that it buys are

$$V_t^n(a,e) = \sigma_\varepsilon \ln\left[\exp\left(\frac{V_t^{rent}(a,e)}{\sigma_\varepsilon}\right) + \exp\left(\frac{V_t^{buy}(a,e)}{\sigma_\varepsilon}\right)\right], \tag{32}$$

$$g_t(a,e) = \frac{\exp\left(V_t^{buy}(a,e)/\sigma_\varepsilon\right)}{\exp\left(V_t^{rent}(a,e)/\sigma_\varepsilon\right) + \exp\left(V_t^{buy}(a,e)/\sigma_\varepsilon\right)}. \tag{33}$$

The owner's expected value and the probability that it sells, $f_t(a,m,r^c,e)$, have the same form with V_t^{own} and V_t^{sell} in place of V_t^{buy} and V_t^{rent}. As $\sigma_\varepsilon \to 0$ the household picks whichever option has the higher value.

A.2 Labor packer and unions

A competitive labor packer buys employment services $L_{k,t}$ from unions $k \in [0,1]$ at wages $W_{k,t}$ and sells aggregate labor L_t to firms,

$$\max_{L_{k,t}} W_t L_t - \int_0^1 W_{k,t} L_{k,t}\, dk \qquad \text{s.t.} \quad L_t = \left[\int_0^1 L_{k,t}^{\frac{\epsilon_w - 1}{\epsilon_w}}\, dk\right]^{\frac{\epsilon_w}{\epsilon_w - 1}}, \tag{34}$$

which gives the demand $L_{k,t} = L_t (W_{k,t}/W_t)^{-\epsilon_w}$ and the wage index

$$W_t = \left[\int_0^1 W_{k,t}^{1-\epsilon_w}\, dk\right]^{\frac{1}{1-\epsilon_w}}.$$

Each union aggregates the efficiency units of its members into a union-specific task, $L_{k,t} = \int e_{i,t}\, \ell_{i,k,t}\, di$, calls on all of them to supply the same hours, $\ell_{i,k,t} = \ell_t$, and resets its wage at random intervals to maximize the utility of its members, as in Erceg et al. (2000) and Schmitt-Grohe and Uribe (2005). Following Auclert et al. (2021a), the union's objective is chosen so that the wage Phillips curve takes the form (27).

A.3 Aggregate household variables

Let $\Psi^h = (a,m,r^c,e)$ and $\Psi^n = (a,e)$ be the states of owners and non-owners, μ_t^h and μ_t^n their distributions, and g_t and f_t the buying and selling probabilities. The aggregates that

enter the rest of the model are

$$
\begin{aligned}
C_t &= \int c_t(\Psi^h)\,\mu_t^h(d\Psi^h) + \int c_t(\Psi^n)\,\mu_t^n(d\Psi^n),\\
M_t^{\text{keep}} &= \int m\,(1-f_t(\Psi^h))\,\mu_t^h(d\Psi^h), \qquad M_t^{\text{buy}} = \int m_t'(\Psi^n)\,g_t(\Psi^n)\,\mu_t^n(d\Psi^n),\\
H_t^{\text{keep}} &= \int (1-f_t(\Psi^h))\,\mu_t^h(d\Psi^h), \qquad H_t^{\text{buy}} = \int g_t(\Psi^n)\,\mu_t^n(d\Psi^n),\\
H_t^r &= \int (1-g_t(\Psi^n))\,\mu_t^n(d\Psi^n) + \int f_t(\Psi^h)\,\mu_t^h(d\Psi^h),\\
A_t' &= \int a_t'(\Psi^h)\,\mu_t^h(d\Psi^h) + \int a_t'(\Psi^n)\,\mu_t^n(d\Psi^n),
\end{aligned}
$$

together with the balance-weighted contract rate of the portfolio,

$$
r_t^c = \frac{\int r^c\, m\,\mu_t^h(d\Psi^h)}{\int m\,\mu_t^h(d\Psi^h)},
$$

and the cash flow $\mathcal{C}_t$ the financial intermediary receives: interest and amortization from owners who keep their house and balances repaid at par by sellers and estates.

A.4 Pricing new loans and valuing the portfolio

Write $p_t \equiv r_t^{\text{new}} - \bar{r}^m$ for the premium of the new-loan rate over the steady-state mortgage rate and $\delta_t \equiv r_t^d - \bar{r}^m$ for the premium of the required return. A loan carries its coupon until it resets, is repaid on sale, or its owner dies, and the financial intermediary discounts every payment along the path of required returns. To first order, the market value of the portfolio at the end of t is

$$
V_t = M_t' + \frac{1}{\xi}\left[\sum_{s=1}^{t} c_{t+1-s}\,p_s + \sum_{k\geq 0}\left(A_k^d\,\delta_{t+k} + \left(A_k - A_k^d\right)p_{t+k}\right)\right], \qquad Q_t = \frac{V_t}{M_t'}. \tag{35}
$$

The first sum is the value of the coupon premia locked in by past originations and resets: balances whose coupon was set at date s carry p_s until they reset, sell or die. The second sum is the effect on the whole portfolio of discounting at the required return rather than at $\bar{r}^m$ (weights A_k^d) and of future resets to the then-prevailing coupon (weights $A_k - A_k^d$). The weights are computed once from the steady-state distribution of owners, summed over the two discount-factor types. Let x index owner states, $m(x)$ the balance, $f(x)$ the probability of selling, and Λ the transition operator of owners who stay (survival, no sale, amortization, idiosyncratic transitions). The present value per unit of balance of a unit coupon premium

ω	w_0	$\sum_{k=0}^{3} w_k$	Mean horizon (quarters)	$Q_1 - 1$ (%)	Δr_1^{new} (bp)	$\Delta Q_1 / \Delta r_1^{\text{new}}$
0.02	0.060	0.23	12.7	-0.59	14.9	-4.0
0.05	0.085	0.31	9.0	-0.57	21.0	-2.7
0.10	0.129	0.43	5.9	-0.52	30.7	-1.7
0.25	0.269	0.72	2.5	-0.35	52.2	-0.7
0.50	0.511	0.95	0.9	-0.16	67.5	-0.2
1.00	1.000	1.00	0.0	-0.00	69.9	–

Table A.1: Par-coupon weights and the portfolio loss per unit of new-loan rate

Notes: Weights at the US calibration for each reset probability ω; the paper uses 0.02 (FRM) and 1 (VRM). The last three columns are the impact responses of the portfolio price and the new-loan rate to the 100 bp tightening without the accelerator, and their ratio in % of par per 100 bps.

on a loan whose owner is in state x next quarter, $\varphi(x)$, solves

$$(1 + \bar{r}^m)\,\varphi(x) = 1 + \xi\,(1 - f(x))\,(1 - \gamma)\,(1 - \omega)\,\mathbb{E}\left[\varphi(x') \mid x\right], \tag{36}$$

since the premium is lost on sale, death or reset. With $\Phi(x) = m(x)\varphi(x)$, ν_k the distribution over owner states of balances whose coupon was set k quarters earlier and has not reset (the rate-exposure density), and β_k the discounted surviving portfolio, $\beta_0 = \mu^h$ and $\beta_{k+1} = \Lambda\beta_k/(1+\bar{r}^m)$, the weights are $c_k = \langle \Phi, \nu_k \rangle$, $A_k^d = -\langle m, \beta_k \rangle/(1+\bar{r}^m)$, and $A_k - A_k^d = \omega\langle \Phi, \beta_k \rangle$ for $k \geq 1$ and zero at $k = 0$, where $\langle \cdot, \cdot \rangle$ sums over states.

New loans are priced at par: their coupon is set so that their value equals the cash lent. With the same weights computed on the landing distribution of the quarter's new loans (superscript buy), the par condition is

$$c^{\text{buy}}\, p_t + \sum_{k \geq 0} A_k^{\text{buy},d}\, \delta_{t+k} + \sum_{k \geq 1} \left(A_k^{\text{buy}} - A_k^{\text{buy},d} \right) p_{t+k} = 0, \tag{37}$$

whose terms are the value of the new loans' coupon premium, the value lost by discounting their payments at a required return above $\bar{r}^m$, and the value of their future resets. Solving (37) backward from the terminal steady state gives the new-loan rate as the weighted average of required returns in (13), with weights w_k that sum to one. Table A.1 reports the weights and the response of the portfolio price to the new-loan rate for several reset probabilities. As ω falls, the weight on the current required return falls, the horizon lengthens, the new-loan rate responds less to a transitory tightening, and the portfolio loses more value per bp of new-loan rate.

The holding return on the portfolio follows from (35) and the par condition as in (14). After the quarter of the shock the portfolio earns exactly the required return, $R_{t+1}^m = r_t^d$, which we verify numerically to a relative precision of 10^{-8}; in the quarter of the shock the

coupon is predetermined and the return carries the revaluation, $R_1^m - \bar{r}^m \approx Q_1 - 1$.

B Solution method

B.1 Steady state

The household problem is solved on a grid of liquid assets, mortgage balances and productivity, with the contract rate at its steady-state value, by an endogenous grid method with an upper envelope to handle the discrete choices and the borrowing limits. The stationary distribution is computed by iterating the transition operator implied by the policy functions and the exogenous processes for productivity, survival and the taste shocks. With the baseline grids the household block has about 1.9 million states. The steady state of the rest of the model is as described in Section 2.3: the portfolio trades at par, all mortgage rates equal $\bar{r} + \iota$, and steady-state leverage divides the fund's claim on the portfolio between deposits and financial intermediary equity.

B.2 The sequence-space system

We study the perfect-foresight response to a one-time unanticipated shock, truncated at $T = 200$ quarters. Given paths for the aggregate inputs of the household block (the wage, hours, transfers, the house price, the rent, the fund return, the new-loan rate and the policy rate, twelve inputs in all once the pieces of the budget constraints are separated), the block returns paths for its aggregate outputs: consumption, savings, the balances of owners who keep and buy, housing demand, purchases, sales, the cash flow to the financial intermediary and the rate-weighted balance that defines the portfolio's coupon. Six aggregate paths are unknown, inflation, the house price, hours, transfers, the required return and the fund return, and six conditions pin them down: the markets for savings, housing, mortgages and labor, the transfer rule, and the fund identity (22). The remaining prices follow from the Phillips curves, the Taylor rule, the par condition, the valuation of the portfolio, the leverage equation and the net-worth law. Writing the six conditions as $H(U, Z) = 0$ for the unknown paths U and the shock Z, the first-order response is $dU = -H_U^{-1} H_Z \, dZ$, where the derivatives of H with respect to the inputs of the household block are its Jacobians, $\mathcal{J}_{t,s} = \partial Y_t / \partial X_s$, stacked with the derivatives of the other blocks (Auclert et al., 2021b). The goods market is not among the six conditions and serves as a check: its first-order residual is below 10^{-5} of the output response in every experiment reported in the paper. In the no-accelerator economies the required return is pinned at $r_{t+1} + \iota$ and the mortgage-market condition determines the deposits the financial intermediary issues rather than a price.

B.3 Household Jacobians

The household Jacobians are computed by the recursion of Bhandari et al. (2023). The method differentiates the household's optimality conditions and the law of motion of the distribution at the steady state, holding fixed at each state the discrete choice (rent or buy, keep or sell) and the set of binding constraints (the loan-to-value and payment-to-income limits and the borrowing limit on liquid assets). The response of the block to news about an input s quarters ahead then follows a linear recursion in s with matrices computed once from the steady-state policies and distribution, which yields the same Jacobian that re-solving the household problem under perturbed price paths would give, without re-solving. We verified this correspondence against Jacobians obtained by finite differences on perturbed paths, each of which requires a full backward pass of the household problem for every input and horizon. The finite-difference Jacobians are also biased wherever a constraint moves with prices, since the perturbation changes which households are constrained; the payment-to-income limit is the clearest case, as it moves six hundred times as much as the new-loan rate per unit of the shock.

The contract rate is a state of each owner, and in the fixed-rate economy it differs across owners. Rather than adding a grid for it, the recursion tracks the rate-exposure densities ν_k of Appendix A.4: the distribution over owner states of balances whose rate was set k quarters earlier and has not reset. A change in the new-loan rate at date s changes the payments of the loans originated or reset at s until they reset, are repaid on sale, or their owner dies, and the response of the block to the new-loan-rate path is the sum over these cohorts, each weighted by its exposure and by the marginal responses of the owners in it. The same densities give the portfolio's coupon, which the financial intermediary block uses in the holding return. In the variable-rate economy every balance resets each quarter, so a single cohort carries the whole exposure and the recursion reduces to the standard one.

C Additional results

C.1 Financial intermediary variables

Figure C.1 shows the financial intermediary variables after the monetary tightening in the four economies at the estimates, including leverage and the portfolio price, which Figure 7 omits. The fixed-rate portfolio loses 1.1% of its value on impact with the accelerator and 0.6% without it; leverage moves by 0.1%. Figure C.2 shows the same variables with the constrained financial intermediary of Section 6.3, where leverage rises 1.7% on impact and the spread reaches 134 bps.

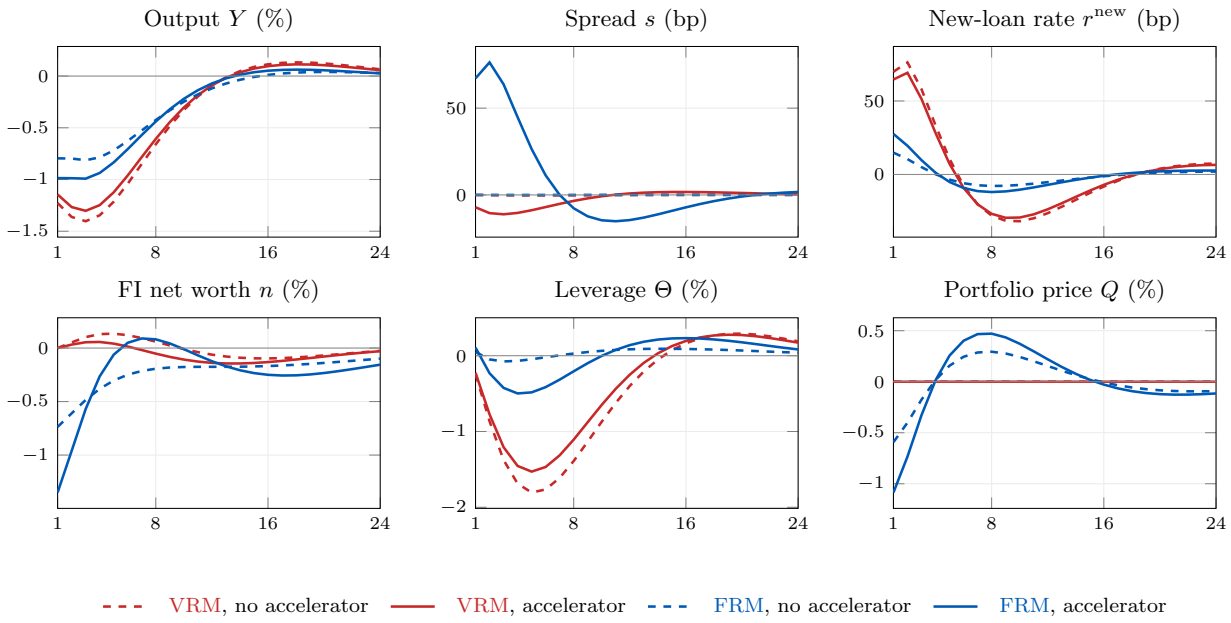


Figure C.1: Financial intermediary variables after the monetary tightening, economy-specific accelerator

Notes: Responses to the 100 bp tightening. Output, net worth, leverage and the portfolio price in % deviations from steady state; the spread and the new-loan rate in annualized bps.

C.2 Productivity and discount-factor shocks

Figures C.3 and C.4 show the responses to the 1% fall in productivity and the 1% rise in the discount factor in the four economies. After the productivity fall the policy rate rises and the fixed-rate spread rises by 10 to 16 bps with it; after the discount-factor rise the policy rate falls, the fixed-rate spread falls, and the portfolio gains value.

C.3 The qualifying-rate counterfactual

Tables C.1 and C.2 complete Section 4.4. The first reports purchases and house prices with the payment-to-income limit applied at the current new-loan rate and at the steady-state rate. The second decomposes the effect of the qualifying rate on consumption into a direct part, the response of the household block to the change in the limit on the baseline path of the new-loan rate, and a general-equilibrium part, the response to the other paths that change.

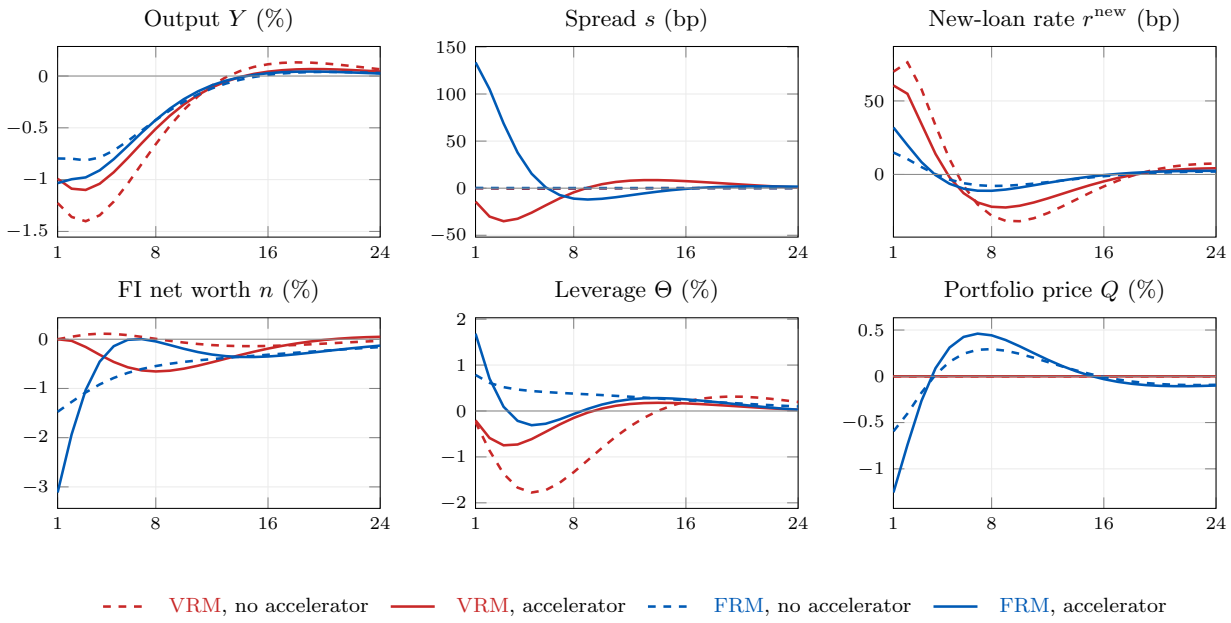


Figure C.2: Financial intermediary variables after the monetary tightening, constrained financial intermediary

Notes: As in Figure C.1, with steady-state leverage 8 and leverage sensitivities at a quarter of the estimates.

	Purchases, t=1 (%)		Purchases, q1–4		p^h, t=1 (%)		p^h, t=4 (%)	
PTI at	r_t^{new}	$\bar{r}^m$	r_t^{new}	$\bar{r}^m$	r_t^{new}	$\bar{r}^m$	r_t^{new}	$\bar{r}^m$
VRM, no FA	−12.20	2.70	−92.4	−7.2	−1.96	−2.29	−2.02	−1.14
VRM, FA (VRM fit)	−11.31	2.56	−83.1	−6.3	−1.82	−2.16	−1.85	−1.09
FRM, no FA	−5.22	−2.06	−25.9	−13.8	−1.19	−1.16	−1.34	−1.17
FRM, FA (FRM fit)	−10.54	−4.97	−49.2	−29.8	−1.32	−1.28	−1.72	−1.45

Table C.1: Purchases and house prices with and without the qualifying-rate channel

Notes: Home purchases in % of their steady-state flow on impact and summed over the first four quarters; the house price in % deviations. FA: the economy-specific accelerator.

D Data

This appendix gives the sources and construction of the two data figures and of the market shares cited in Section 5.1. All series are public. The scripts that download and assemble them are in the replication package (`Empirical/mortgage_rates_passthrough.py` and `Empirical/lender_side_passthrough.py`); the series were downloaded in October 2026.

Figure 1. All series are monthly averages in %.

United States. The policy rate is the effective federal funds rate (FRED series FEDFUNDS). The rate on new mortgages is the Freddie Mac Primary Mortgage Market Survey

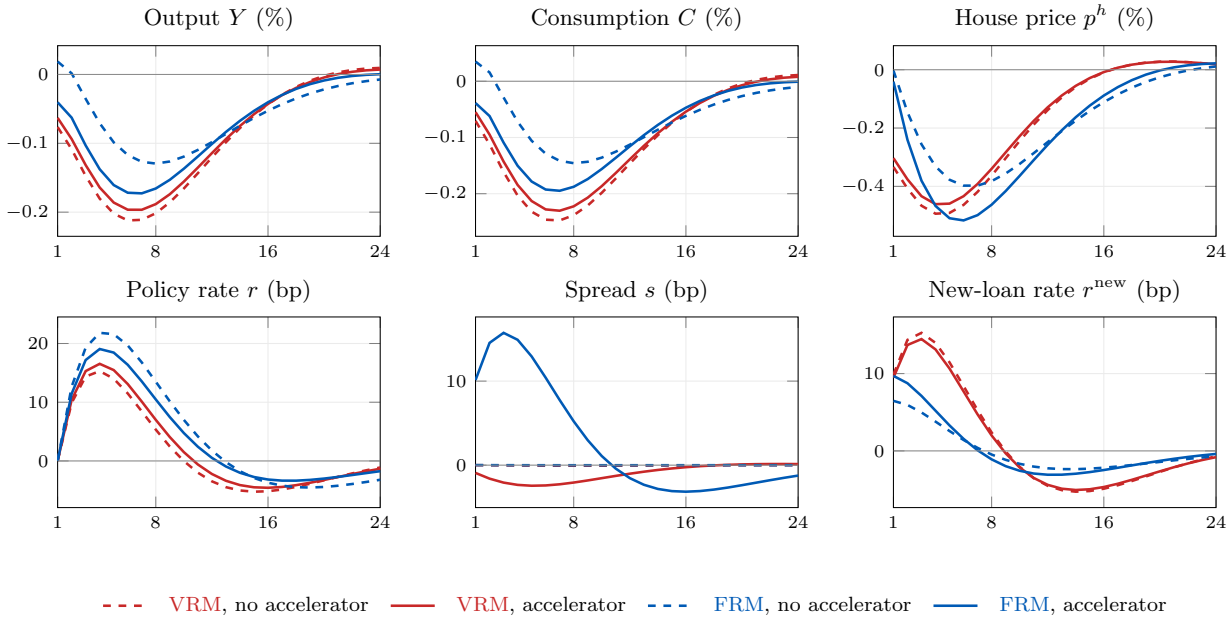


Figure C.3: A 1% fall in productivity

Notes: Productivity falls 1% with persistence 0.79. Output, consumption and the house price in % deviations from steady state; the policy rate, the spread and the new-loan rate in annualized bps.

	Quarters 1–4			Quarters 1–12				
	Total	Direct	GE	Total	Direct	GE	of which p^h	N, w, T
VRM, no FA	−1.27	−1.07	−0.19	−2.85	0.04	−2.89	−1.05	−1.54
VRM, FA (VRM fit)	−1.06	−0.95	−0.11	−2.34	0.11	−2.44	−0.86	−1.31
FRM, no FA	−0.28	−0.10	−0.18	−0.46	0.21	−0.68	−0.29	−0.25
FRM, FA (FRM fit)	−0.42	−0.20	−0.22	−0.66	0.30	−0.96	−0.43	−0.36

Table C.2: The qualifying-rate channel's effect on consumption

Notes: Consumption at the baseline minus consumption with the limit at the steady-state rate, in pp-quarters of steady-state consumption. Direct: the part of the household Jacobian that works through the limit, on the baseline path of the new-loan rate. GE: the response to the other paths that change, split into the house price and the hours, wage and transfer paths.

30-year fixed rate, weekly (FRED MORTGAGE30US), averaged by month. The rate on outstanding mortgages is the average interest rate on outstanding closed-end first-lien mortgages in the Federal Housing Finance Agency's National Mortgage Database aggregate statistics (series AVE_INTRATE, all mortgages, national), quarterly, placed at the last month of each quarter (Federal Housing Finance Agency, 2026).

Canada. The policy rate is the Bank of Canada's target for the overnight rate (Valet series V39079), daily, averaged by month. The rate on new mortgages is the interest rate on funds advanced for residential mortgages, and the rate on outstanding mortgages the interest rate on outstanding residential mortgage balances, both from Statistics Canada

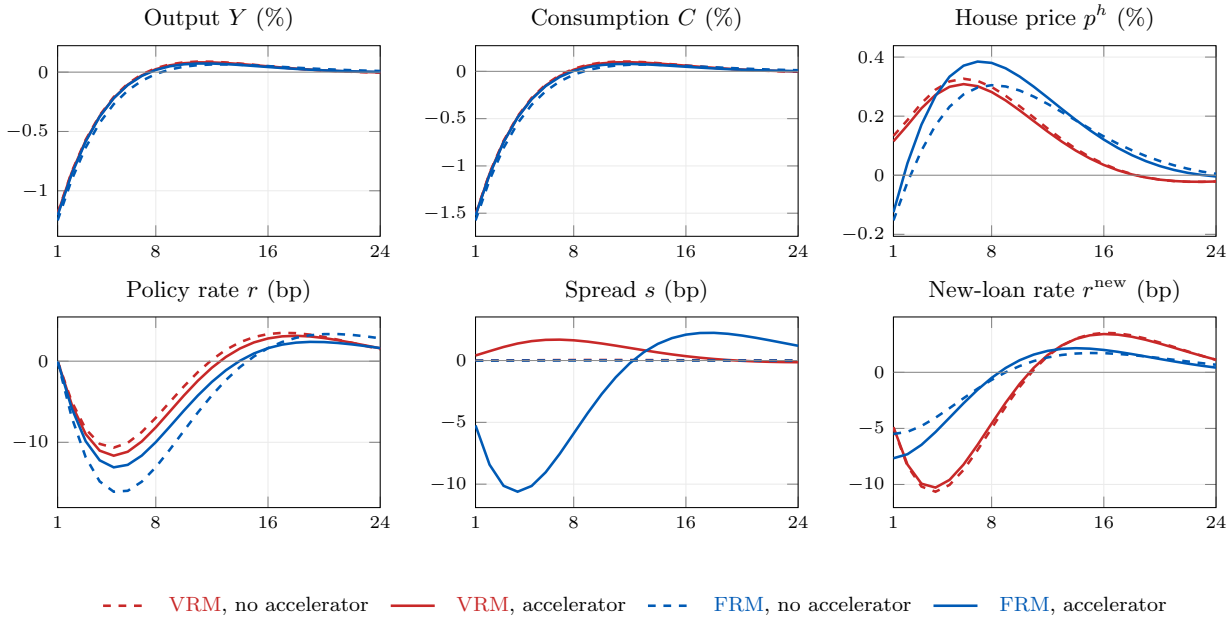


Figure C.4: A 1% rise in the discount factor

Notes: All households' discount factors rise 1% with persistence 0.72. Units as in Figure C.3.

table 10-10-0006-01 (funds advanced, outstanding balances, and interest rates for new and existing lending, chartered banks). Each is the average of the insured and uninsured rates weighted by the corresponding dollar amounts.

United Kingdom. The policy rate is Bank Rate (Bank of England series IUMABEDR, monthly average). The rates on new and outstanding mortgages are the Bank of England's effective interest rates on sterling loans secured on dwellings to households, new advances (CFMBJ95) and outstanding balances (CFMHSDE).

Australia. The policy rate is the cash rate target (Reserve Bank of Australia table F1.1, series FIRMMCRT, monthly average). The rates on new and outstanding mortgages are the lending rates on owner-occupier housing credit for new loans funded in the month (table F6, FLRHOFTA) and for outstanding loans (FLRHOOTA), all institutions, available from July 2019.

Figure 6. The US mortgage spread is the Freddie Mac 30-year rate minus the 10-year Treasury constant-maturity yield (FRED DGS10), both averaged by month. The spreads of new-mortgage rates over policy rates for Canada and the United Kingdom use the series of Figure 1; for Australia the numerator is the rate on new owner-occupier variable-rate loans (table F6, FLRHOFVA), so that the panel is not affected by the share of fixed-rate loans in new lending. Unrealized gains and losses on investment securities are from the FDIC

Quarterly Banking Profile for the second quarter of 2026, chart data for all FDIC-insured institutions, expressed in % of total assets (Federal Deposit Insurance Corporation, 2026).

Market shares. The shares of outstanding US mortgages by product (30-year fixed, 15-year fixed, adjustable) and by market (acquired by the enterprises, government programs, other conventional) are from the same National Mortgage Database statistics at the fourth quarter of 2024: the product shares are the series PCT_TERM_FRM_30 and PCT_TERM_FRM_15, with the adjustable-rate share the sum of PCT_TERM_ARM_1_4 and PCT_TERM_ARM_5PL. The market shares are the series PCT_ENTERPRISE, PCT_GOVERNMENT, and PCT_OTHER. The variable-rate share of outstanding Canadian mortgage balances is the sum of the insured and uninsured variable-rate balances over the sum of the insured and uninsured total balances in Statistics Canada table 10-10-0006-01.